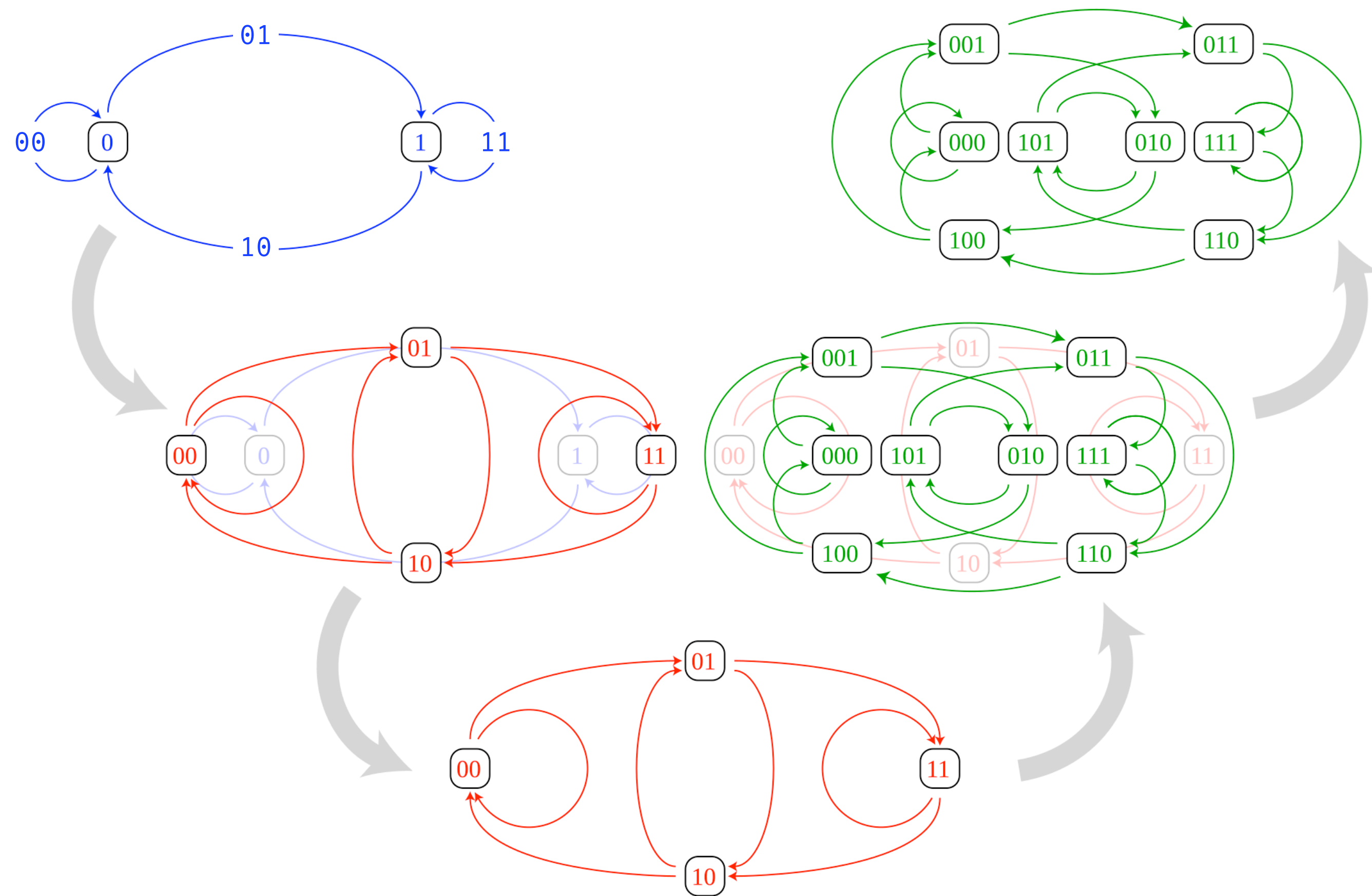


The de Bruijn Graph and its efficient representation

The De Bruijn graph (DBG, dbG) & its variants

Origins

DBG on binary alphabet



Directed graph representing overlaps between sequences of symbols.

Vertices:

$$V = S^n = \{(s_1, \dots, s_1, s_1), (s_1, \dots, s_1, s_2), \dots, (s_m, \dots, s_m, s_m)\}$$

n is the “order” of the DBG

Edges:

$$E = \{((t_1, t_2, \dots, t_n), (t_2, \dots, t_n, s_j)) : t_i \in S, 1 \leq i \leq n, 1 \leq j \leq m\}$$



Nicolaas Govert de Bruijn

(“de brun” or “de brown”;
not “de bru-jin”)

The De Bruijn graph (DBG, dbG) & its variants

Popularized for genome assembly (Note: induced subgraph of *actual* DBG)

An Eulerian path approach to DNA fragment assembly

Pavel A. Pevzner*, Haixu Tang[†], and Michael S. Waterman^{†‡§}

*Department of Computer Science and Engineering, University of California, San Diego, La Jolla, CA; and Departments of [†]Mathematics and [‡]Biological Sciences, University of Southern California, Los Angeles, CA

Contributed by Michael S. Waterman, June 7, 2001

Original Genome:

ATCGATCGTA

Sequencing Reads:

ATCG

TCGA

CGAT

GATC

ATCG

TCGT

CGTA

Unique k-mers (k=3)

ATC, CGT, CGA, TCG, GAT, GTA

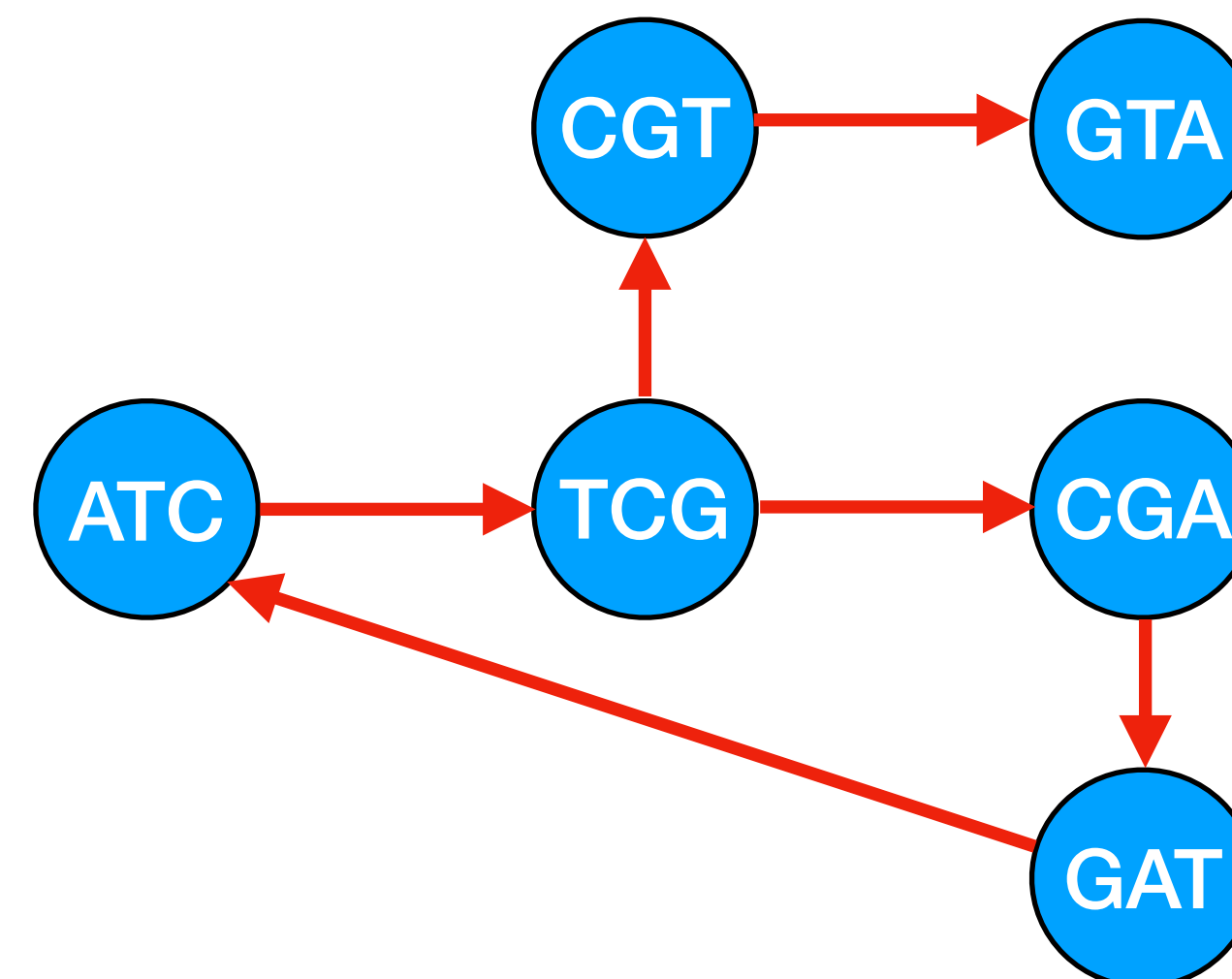
Eulerian Path:

ATC→TCG→CGA→GAT→ATC→TCG→CGT→GTA

Reconstruction:

ATC+G+A+T+C+G+T+A = ATCGATCGTA

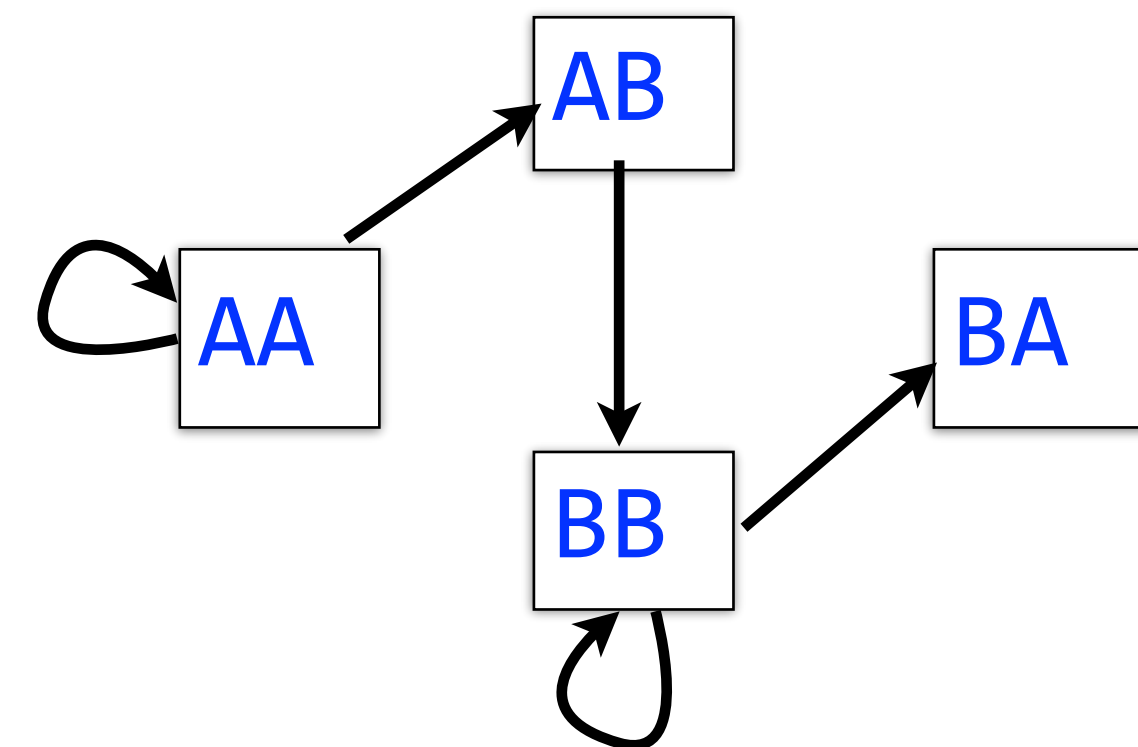
De Bruijn Graph:



- Popularized as sequencing throughput increased.
- Vertices are (k-1)-mers, edges are k-mers
- Replaced expensive inexact read “overlap alignment” with exact matching 2 (k-1)-mers are joined by an edge if the co-occur in the same k-mer
- Vastly improved scalability (time & memory)
- Beautiful theory (in the “perfect” case)
- Works well in practice (with appropriate heuristics)

Eulerian walk definitions and statements

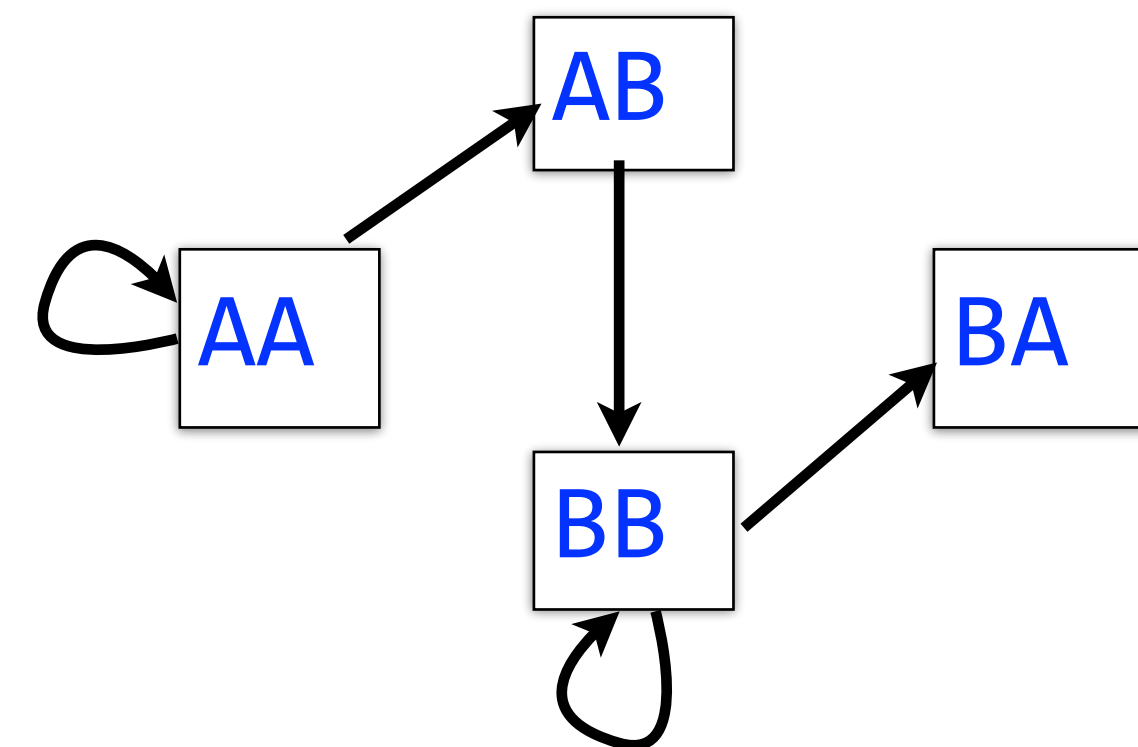
Node is balanced if indegree equals outdegree



Eulerian walk definitions and statements

Node is balanced if indegree equals outdegree

Node is semi-balanced if indegree differs from outdegree by 1

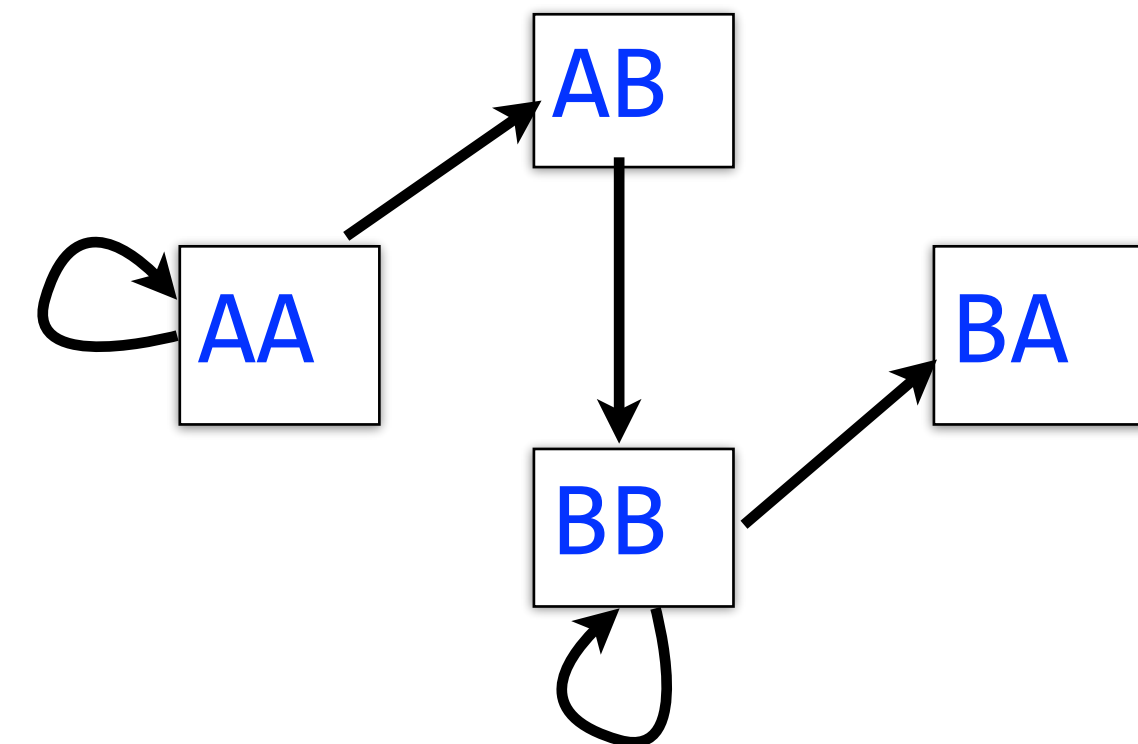


Eulerian walk definitions and statements

Node is balanced if indegree equals outdegree

Node is semi-balanced if indegree differs from outdegree by 1

Graph is connected if each node can be reached by some other node



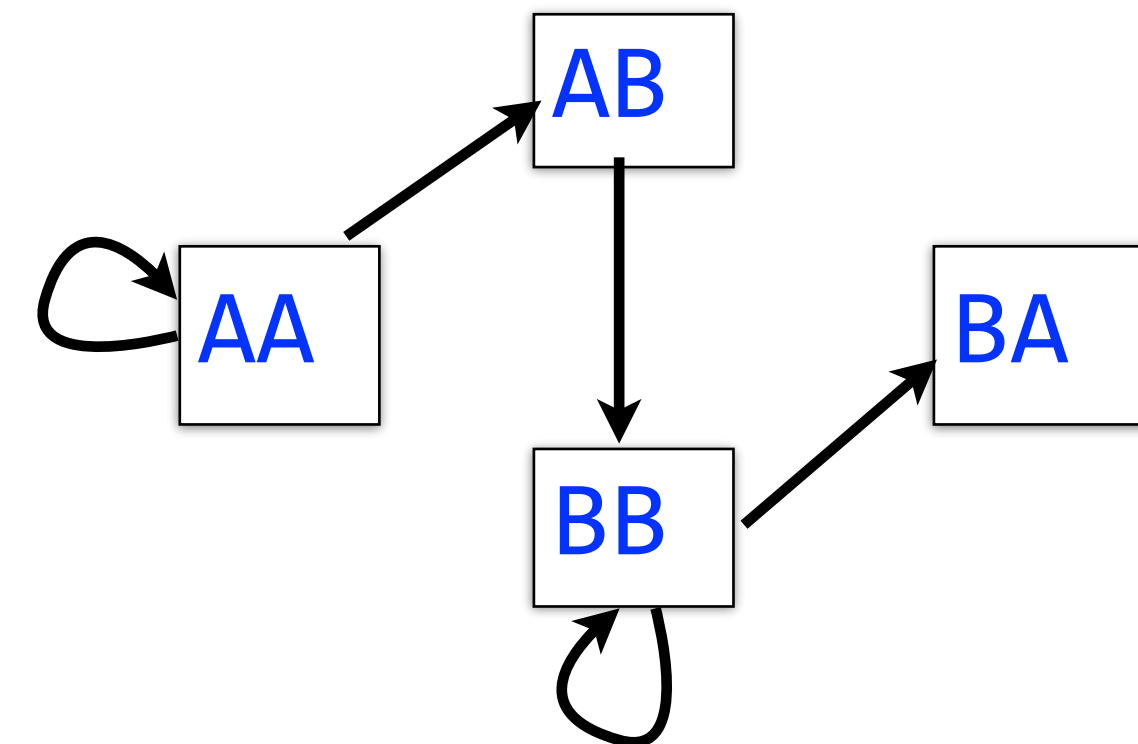
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Eulerian walk definitions and statements

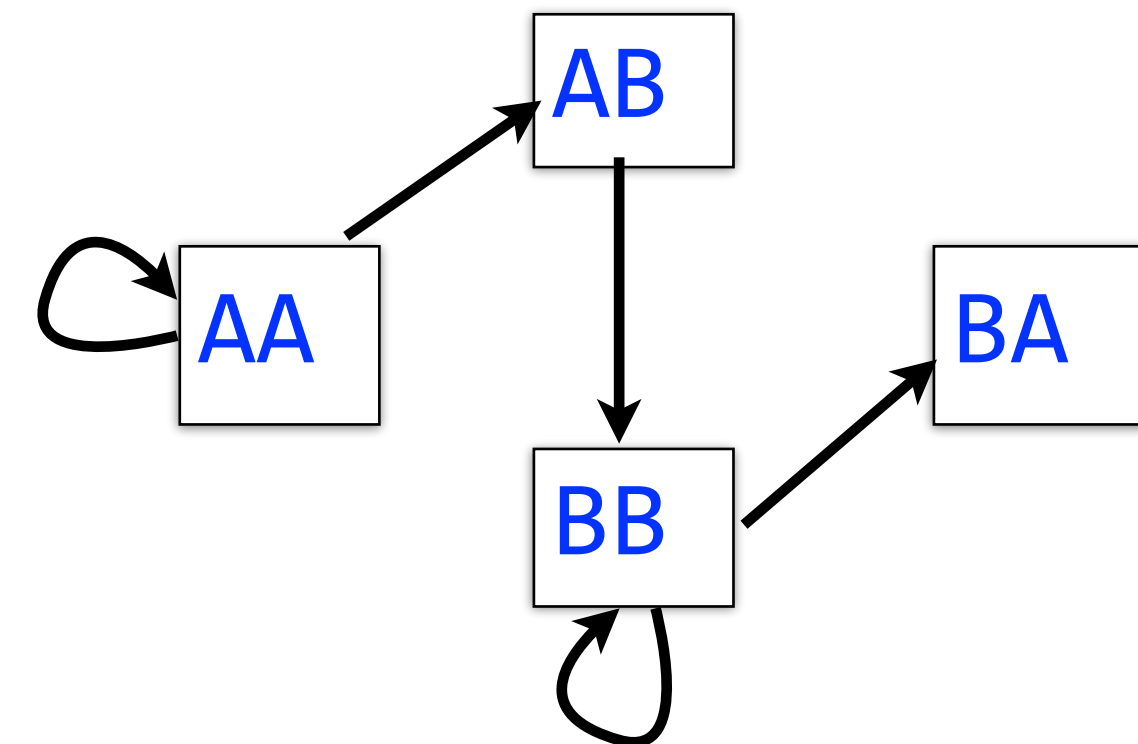
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Not all graphs have Eulerian walks. Graphs that do are Eulerian. (For simplicity, we won't distinguish Eulerian from semi-Eulerian.)



Eulerian walk definitions and statements

Node is balanced if indegree equals outdegree

Node is semi-balanced if indegree differs from outdegree by 1

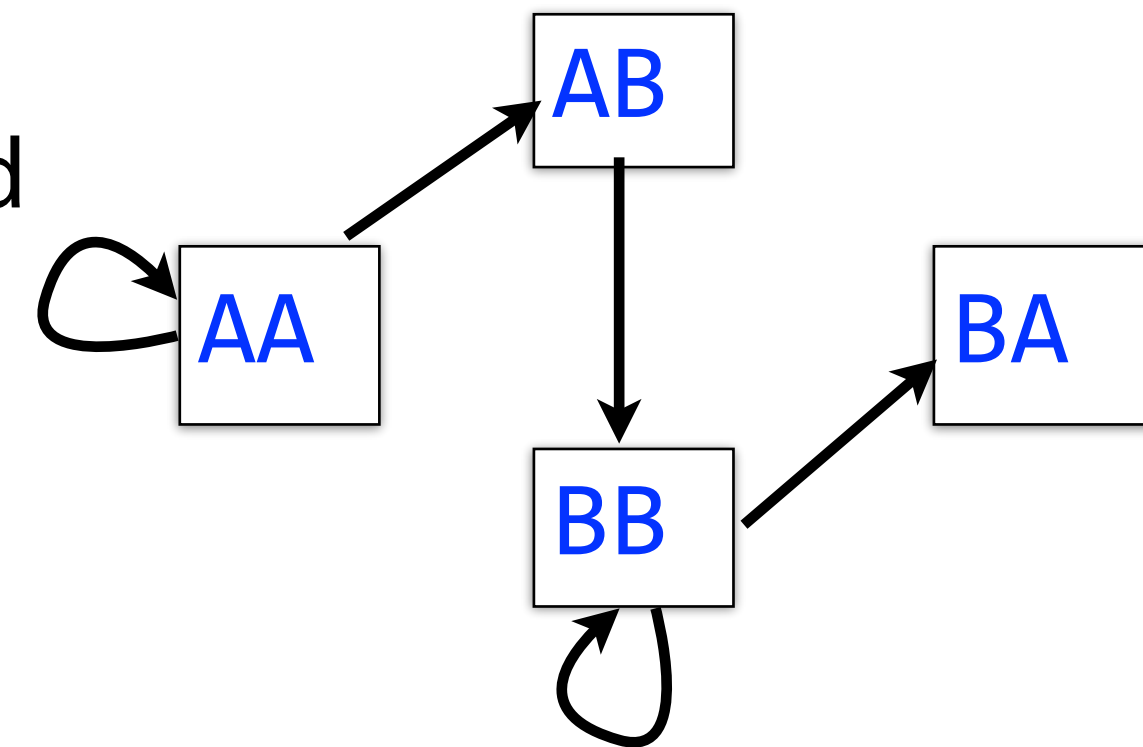
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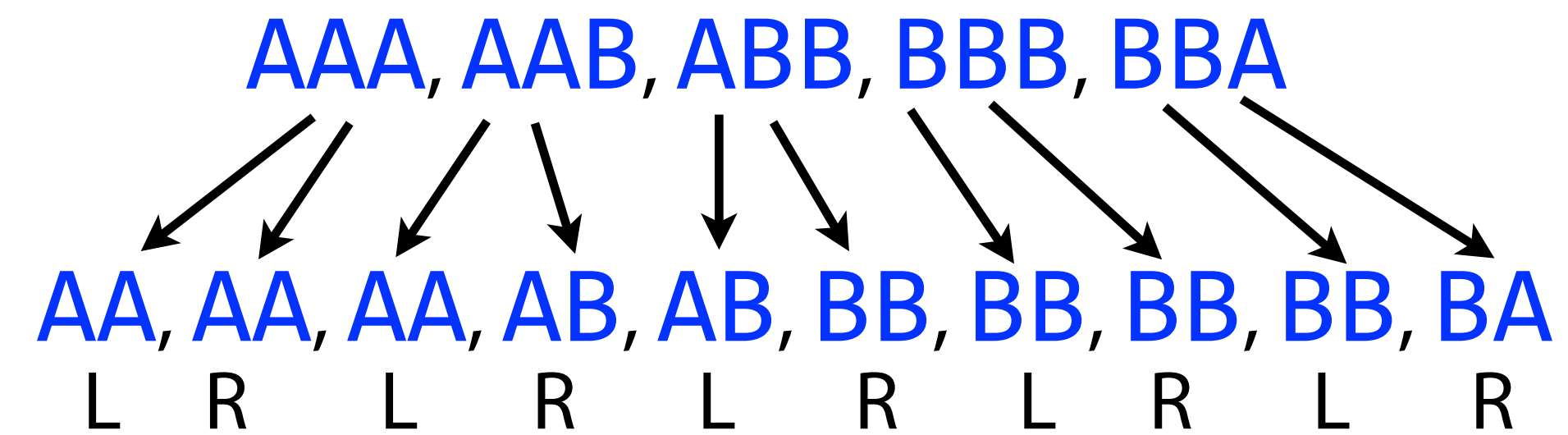
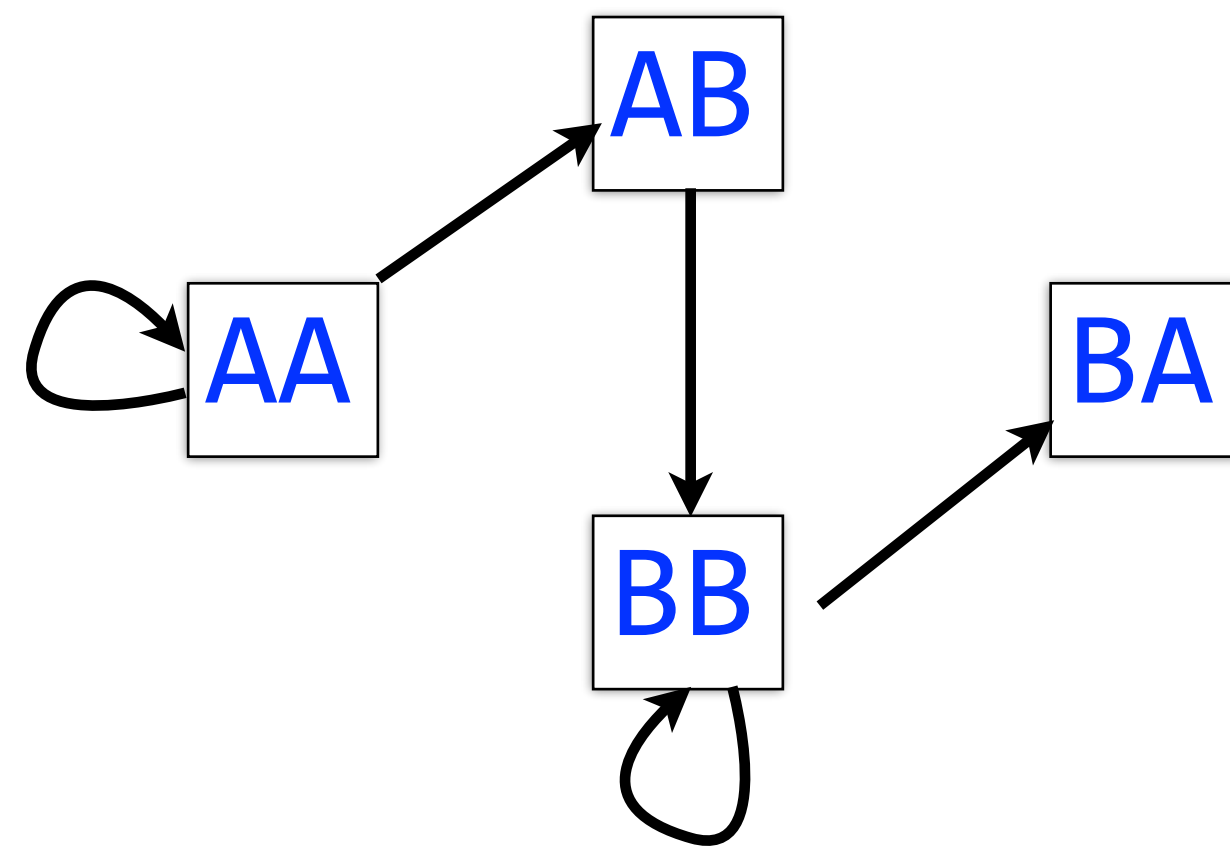
A directed, connected graph is Eulerian if and only if it has at most 2 semi-balanced nodes and all other nodes are balanced

Jones and Pevzner section 8.8



De Bruijn graph

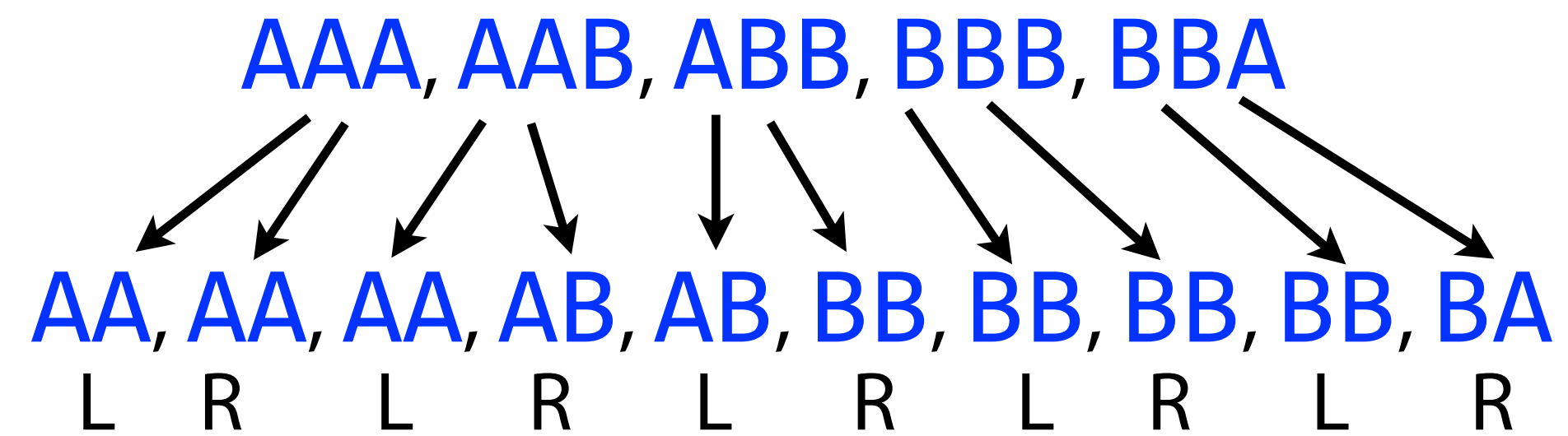
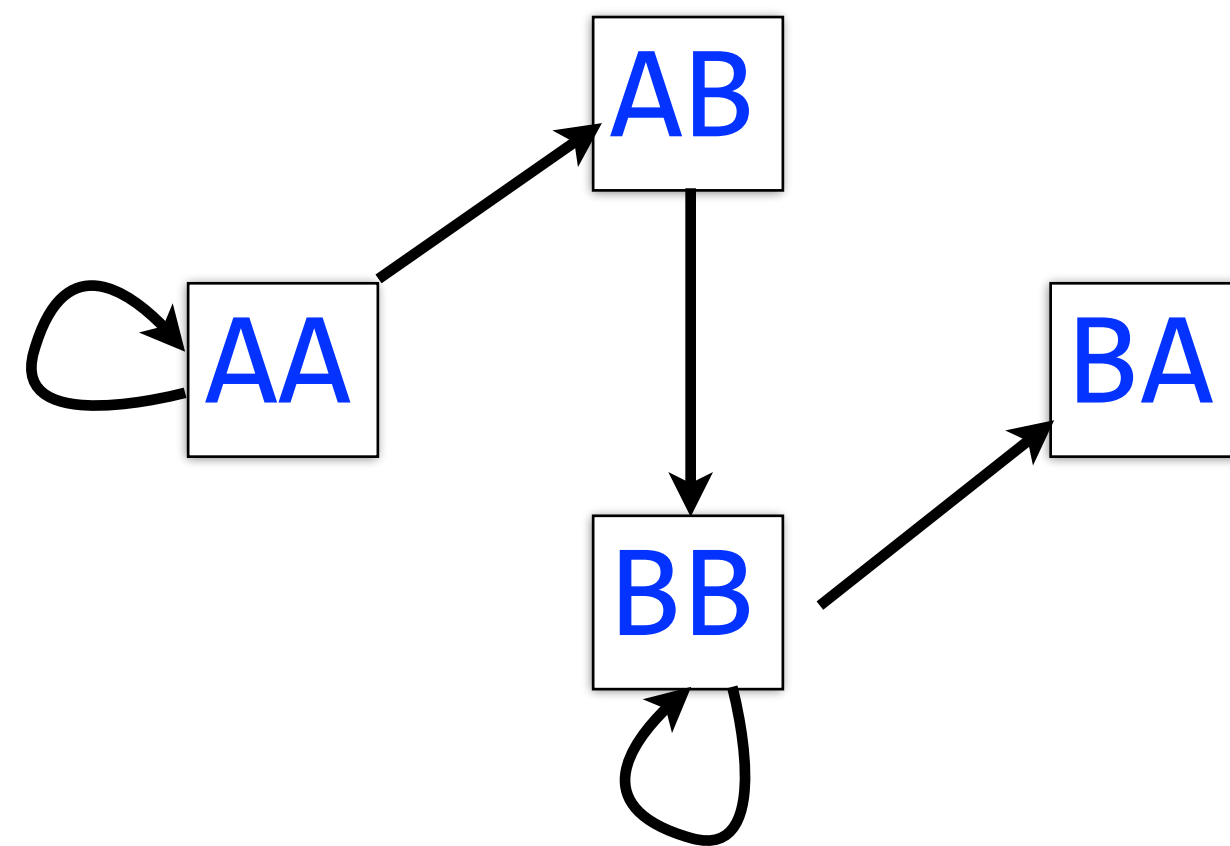
Back to de Bruijn graph



Is it Eulerian?

De Bruijn graph

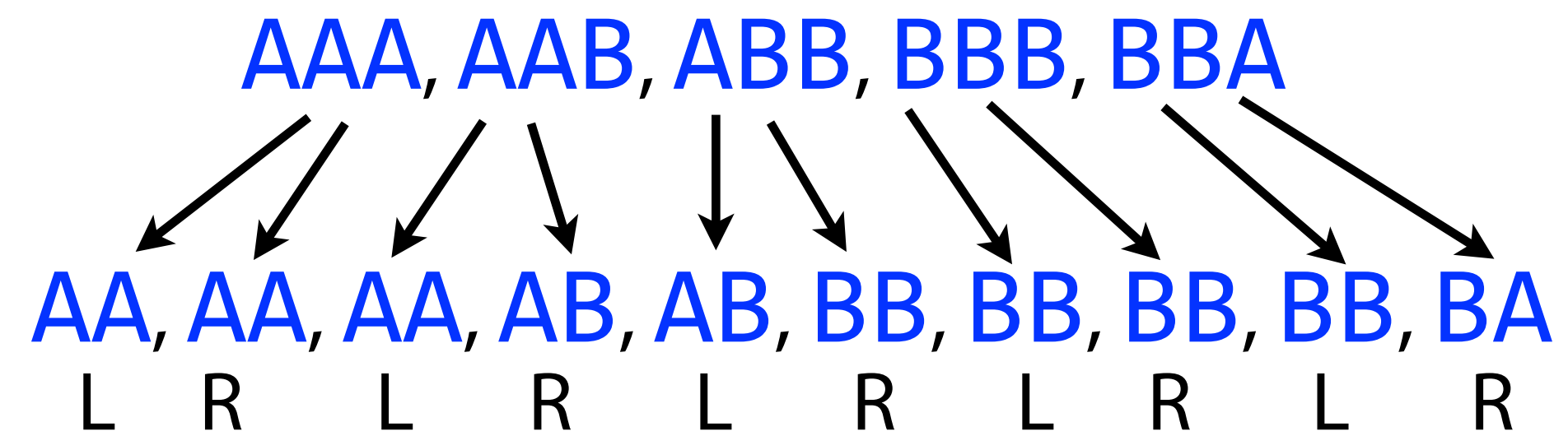
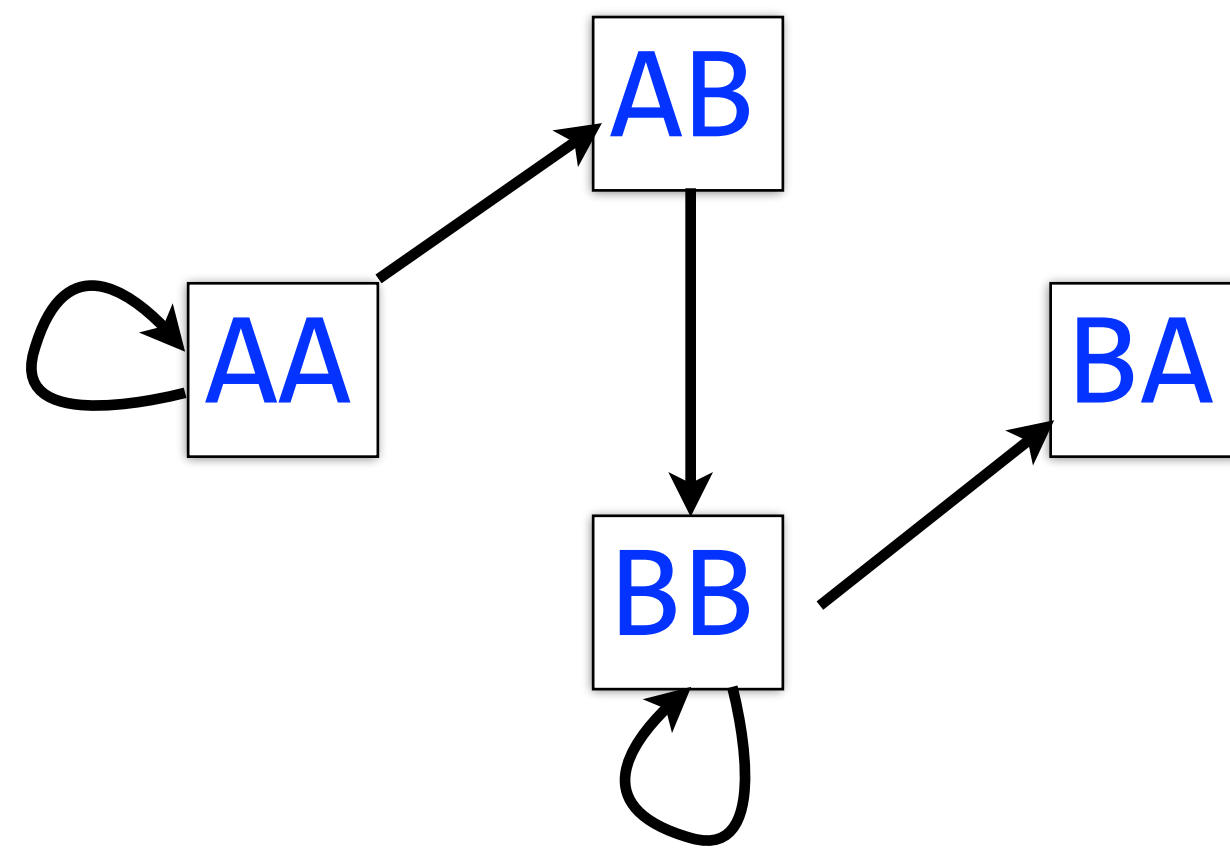
Back to de Bruijn graph



Is it Eulerian? Yes

De Bruijn graph

Back to de Bruijn graph

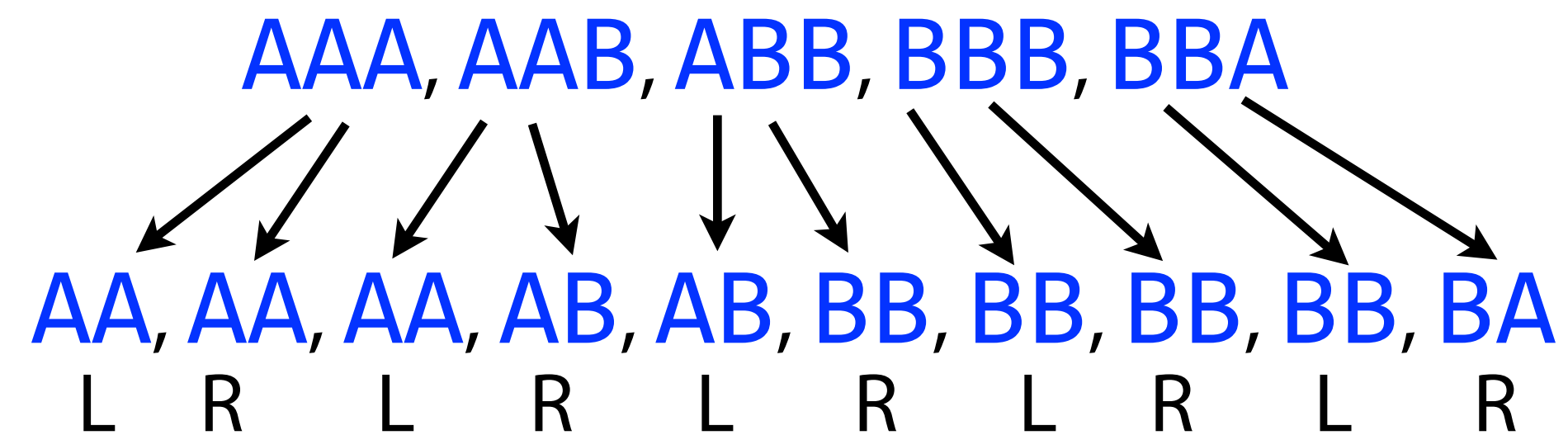
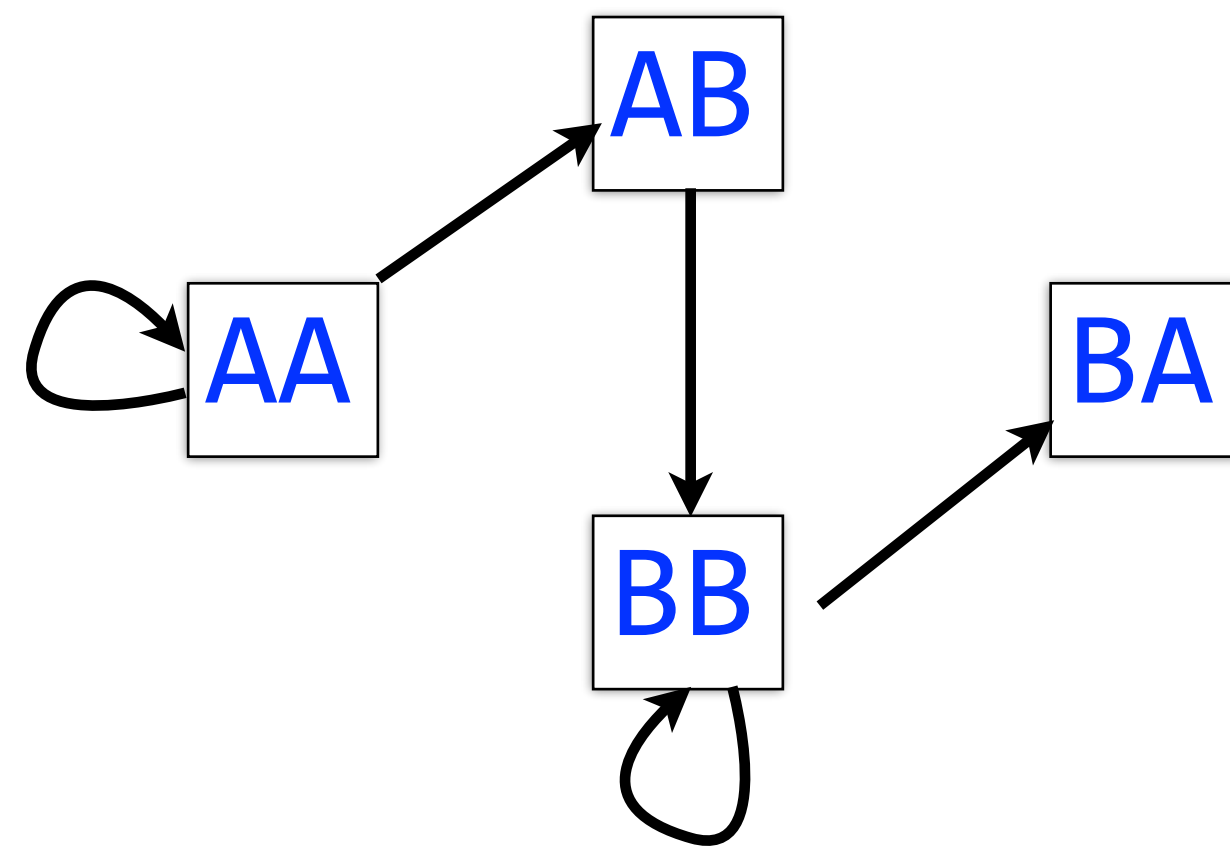


Is it Eulerian? Yes

Argument 1: AA → AA → AB → BB → BB → BA

De Bruijn graph

Back to de Bruijn graph



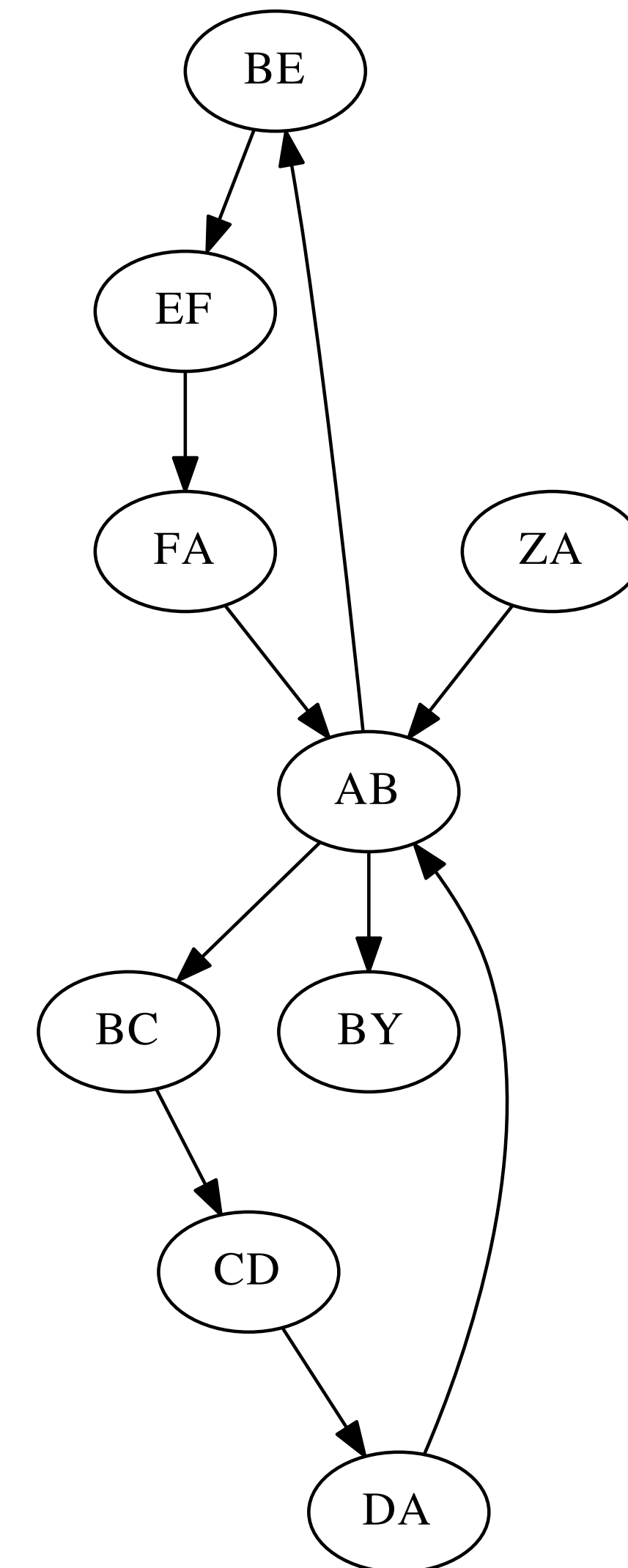
Is it Eulerian? Yes

Argument 1: $AA \rightarrow AA \rightarrow AB \rightarrow BB \rightarrow BB \rightarrow BA$

Argument 2: AA and BA are semi-balanced, AB and BB are balanced

De Bruijn graph

Problem 1: Repeats still cause misassemblies

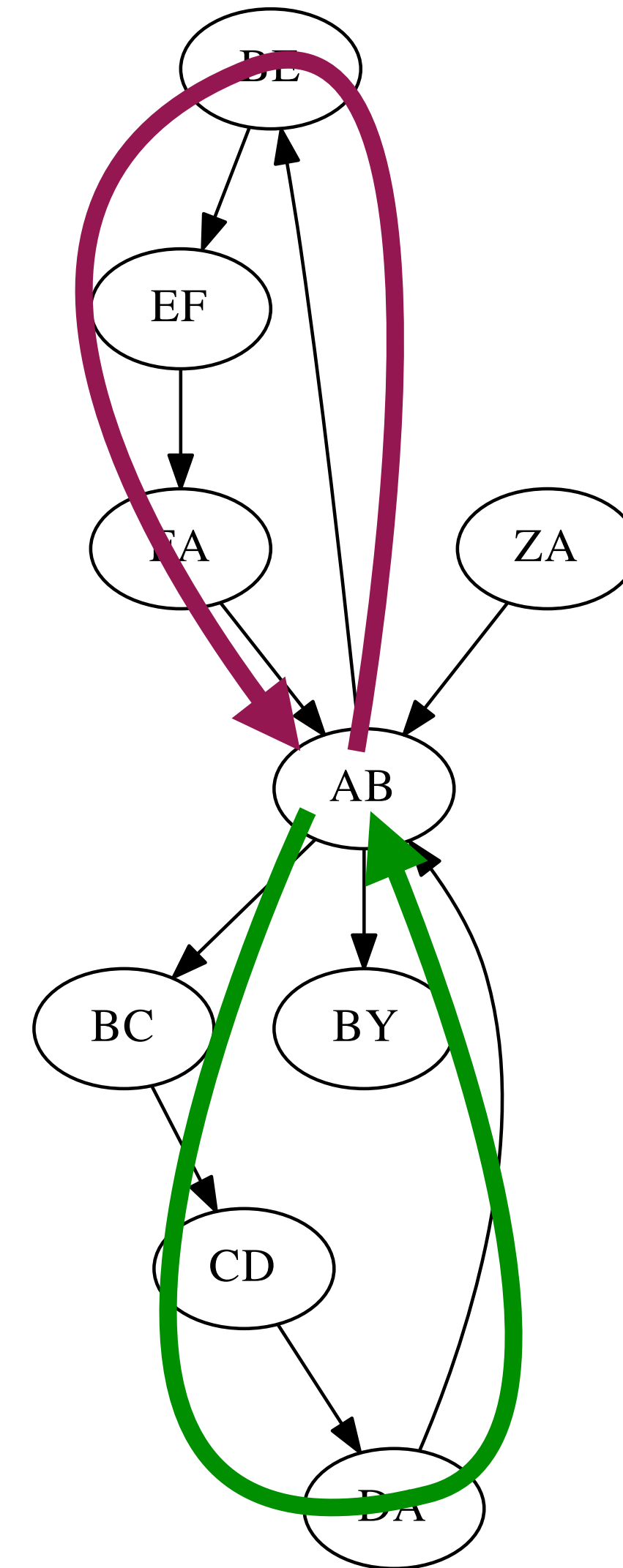


De Bruijn graph

Problem 1: Repeats still cause misassemblies

ZA → AB → BE → EF → FA → AB → BC → CD → DA → AB → BY

ZA → AB → BC → CD → DA → AB → BE → EF → FA → AB → BY



De Bruijn graph

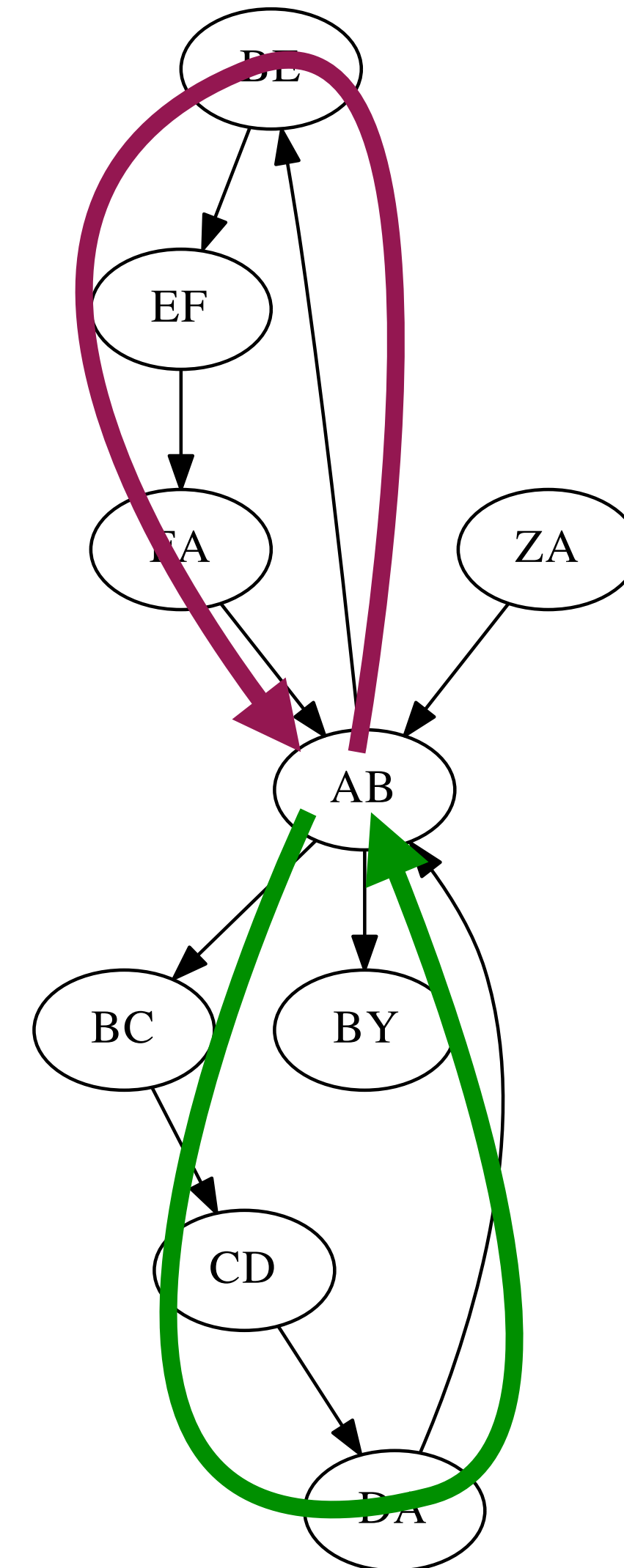
Problem 1: Repeats still cause misassemblies

ZA → AB → BE → EF → FA → AB → BC → CD → DA → AB → BY

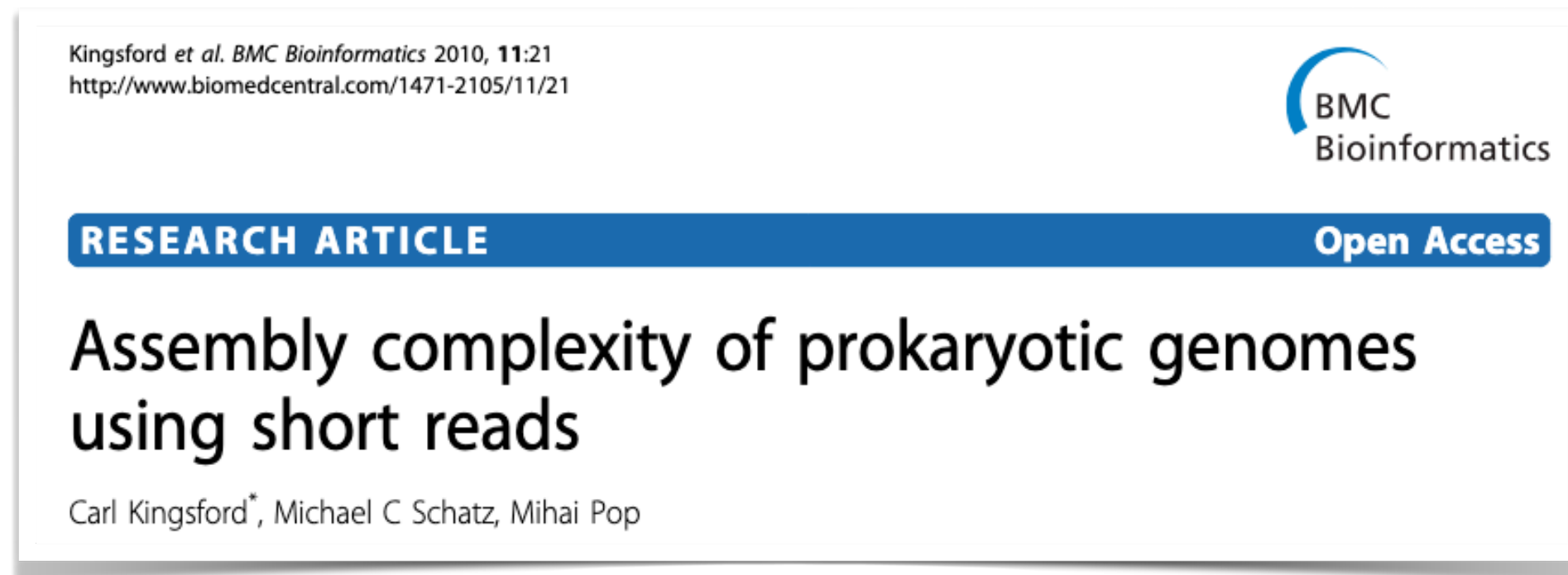
ZA → AB → BC → CD → DA → AB → BE → EF → FA → AB → BY

Problem 2:

We've been building DBGs assuming "perfect" sequencing: each k-mer reported exactly once, no mistakes. Real datasets aren't like that.



You can find *a* path; not necessarily *the* path



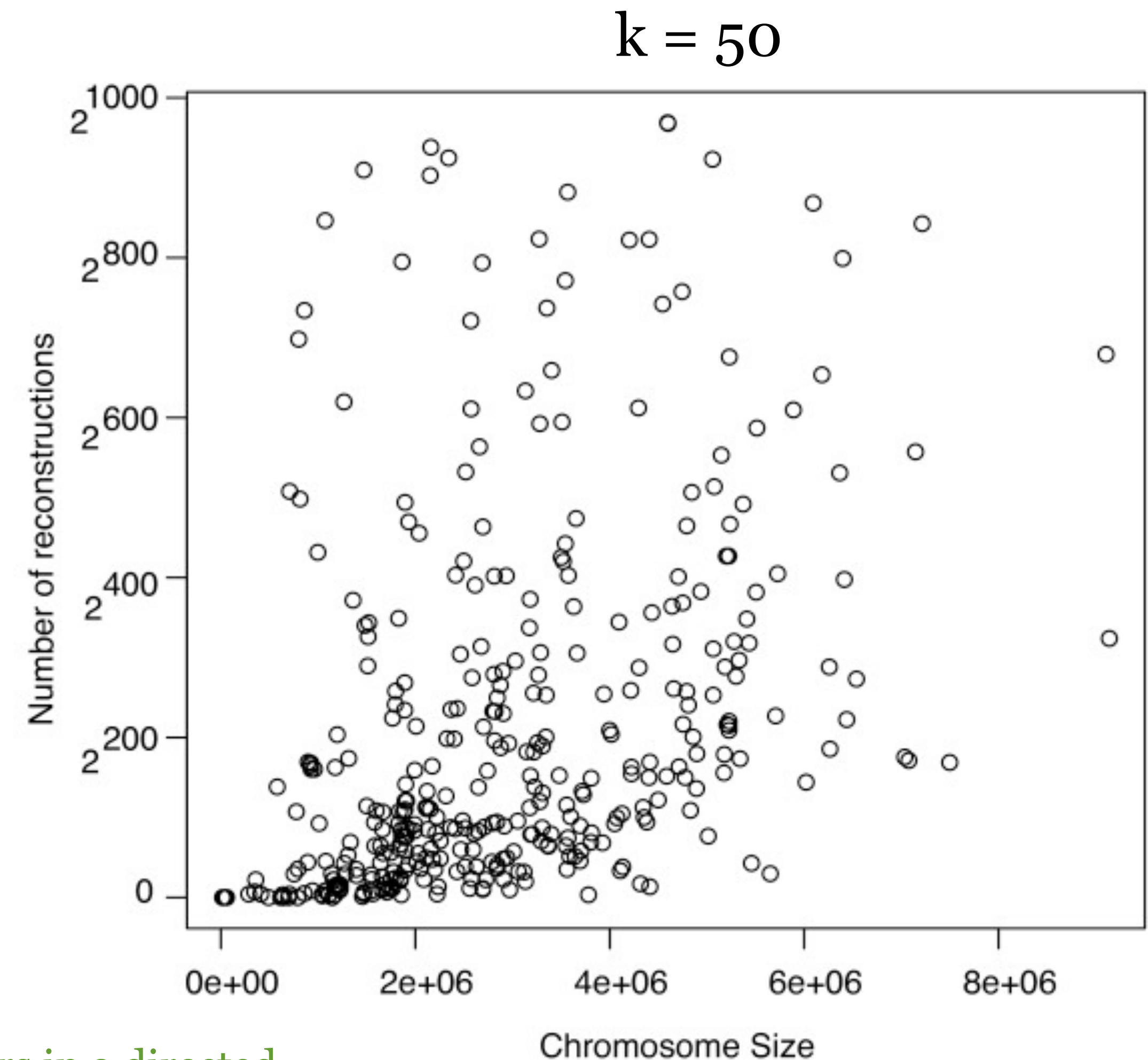
There are typically an astronomical number of possible Eulerian tours with perfect data.

Adding back constraints to limit # of tours leads to a NP-hard problem.

With imperfect data, there are usually NO Eulerian tours

Estimating # of parallel edges is usually tricky.

Aside: counting # of Eulerian tours in a directed graph is easy, but in an undirected graph is #P-complete (hard).



(Kingsford, Schatz, Pop, 2010)

* slide courtesy of Carl Kingsford

A challenge — de Bruijn graphs can get *big*!

Genomes can contain billions (or more) of true k-mers

Even storing the k-mers one-by-one, explicitly, can become very expensive

Overhead can be even worse if we are explicitly storing graph structure as well

Need to think of *compact* ways to store the de Bruijn graph.

Bloom Filters

Originally designed to answer *probabilistic* membership queries:

Is element e in my set S ?

If yes, **always** say yes

If no, say no **with large probability**

False positives can happen; false negatives cannot.

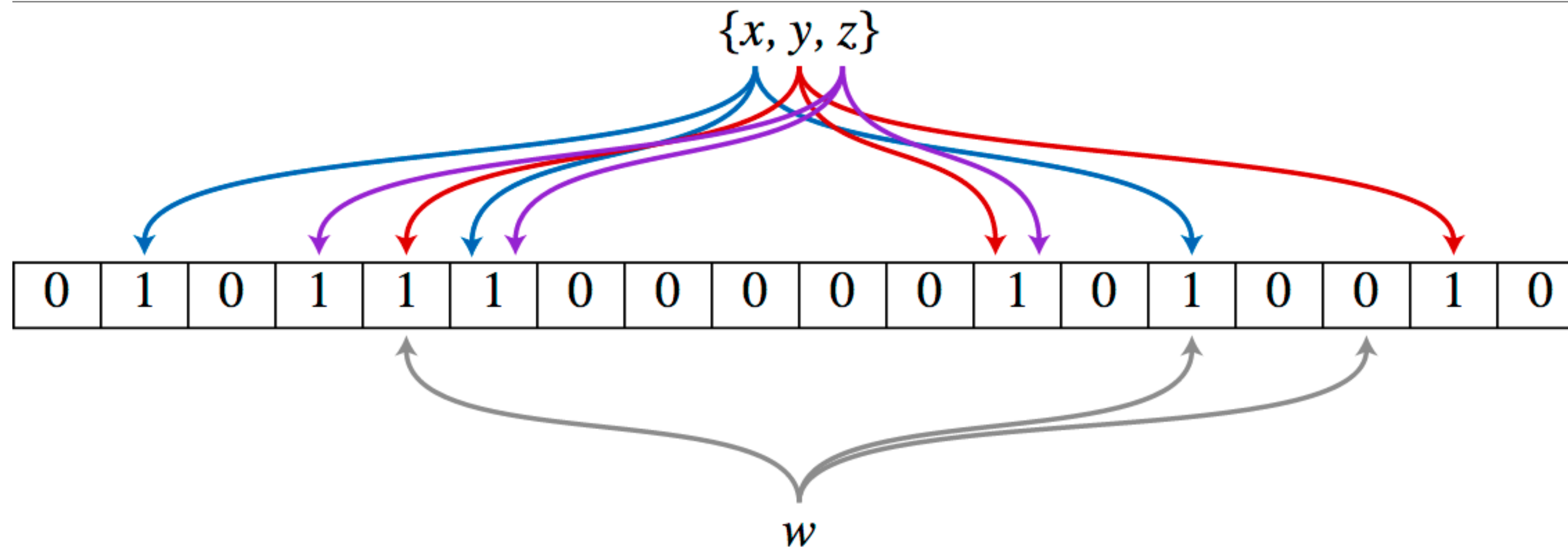
Bloom Filters

For a set of size N , store an array of M bits

Use k different hash functions, $\{h_0, \dots, h_{k-1}\}$

To insert e , set $A[h_i(e)] = 1$ for $0 < i < k$

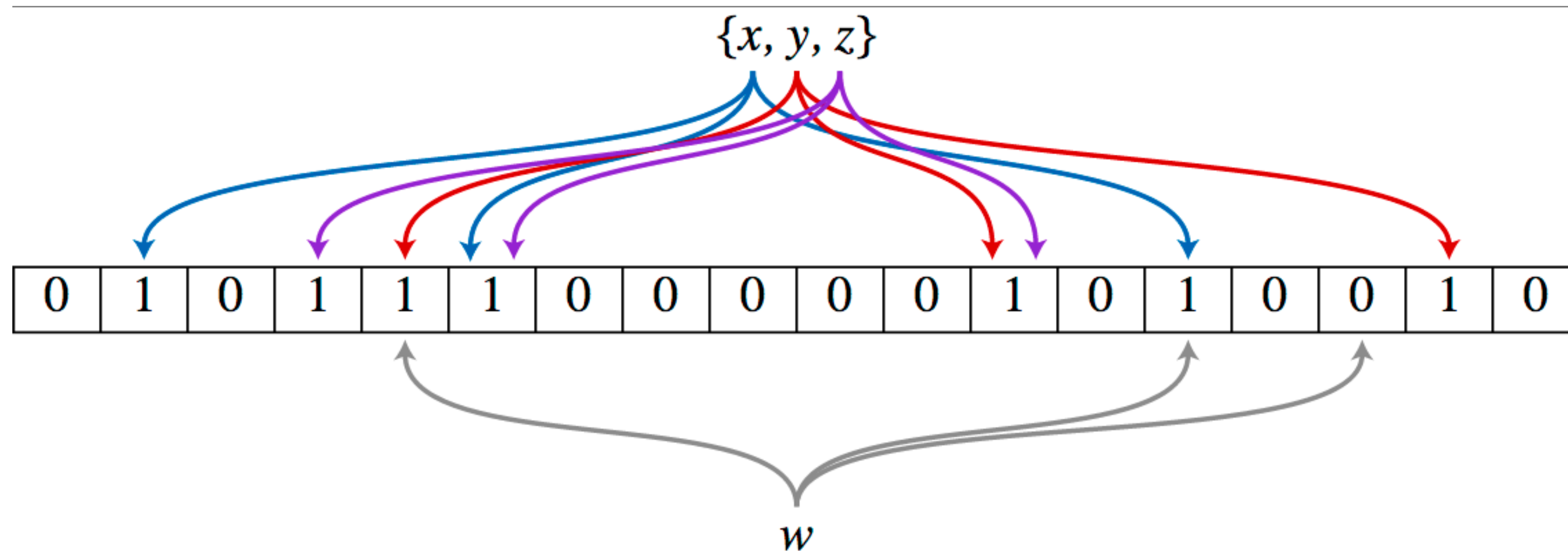
To query for e , check if $A[h_i(e)] = 1$ for $0 < i < k$



Bloom Filters

If hash functions are good and sufficiently independent, then the probability of false positives is low and controllable.

How low?



False Positives

Let q be the fraction of the m -bits which remain as 0 after n insertions.

The probability that a randomly chosen bit is 1 is $1-q$.

But we need a 1 in the position returned by k different hash functions; the probability of this is $(1-q)^k$

We can derive a formula for the expected value of q ,
for a filter of m bits, after n insertions with k different hash functions:

$$E[q] = (1 - 1/m)^{kn}$$

False Positives

Mitzenmacher & Unfal used the Azuma-Hoeffding inequality to prove (without assuming the probability of setting each bit is independent) that

$$\Pr(|q - E[q]| \geq \frac{\lambda}{m}) \leq 2\exp(-2\frac{\lambda^2}{m})$$

That is, the random realizations of q are highly concentrated around $E[q]$, which yields a false positive prob of:

$$\sum_t \Pr(q = t)(1 - t)^k \approx (1 - E[q])^k = \left(1 - \left[1 - \frac{1}{m}\right]^{kn}\right)^k \approx (1 - e^{-\frac{kn}{m}})^k$$

False Positives

$$\sum_t \Pr(q = t)(1 - t)^k \approx (1 - E[q])^k = \left(1 - \left[1 - \frac{1}{m}\right]^{kn}\right)^k \approx (1 - e^{-\frac{kn}{m}})^k$$

This lets us choose optimal values to achieve a target false positive rate. For example, assume m & n are given. Then we can derive the optimal k

$$k = (m/n) \ln 2 \Rightarrow 2^{-k} \approx 0.6185^{m/n}$$

We can then compute the false positive prob

$$p = \left(1 - e^{-\left(\frac{m}{n} \ln 2\right) \frac{n}{m}}\right)^{\left(\frac{m}{n} \ln 2\right)} \implies$$

$$\ln p = -\frac{m}{n} (\ln 2)^2 \implies$$

$$m = -\frac{n \ln p}{(\ln 2)^2}$$

False Positives

$$\sum_t \Pr(q = t)(1 - t)^k \approx (1 - E[q])^k = \left(1 - \left[1 - \frac{1}{m}\right]^{kn}\right)^k \approx (1 - e^{-\frac{kn}{m}})^k$$

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$$\ln p = -\frac{m}{n} (\ln 2)^2 \Rightarrow$$

$$m = -\frac{n \ln p}{(\ln 2)^2}$$

given an **expected # elems** and a **desired false positive rate**

we can compute the **optimal size** and **# of hash functions**

Bloom Filters & De Bruijn Graphs

How could the Bloom filter be useful for representing a De Bruijn graph?

Say we have a bloom filter B , for all of the k -mers in our data set, and say I give you one k -mer that is truly present.

Remember this term, it will come up in a couple of slides

We now have a “*navigational*” representation of the De Bruijn graph (can return the set of neighbors of a node, but not select/iterate over nodes); why?

Detour: Bloom Filters & De Bruijn Graphs

How could this data structure be useful for representing a De Bruijn graph?



A given (k-1)-mer can only have $2 \cdot |\Sigma|$ neighbors;
 $|\Sigma|$ incoming and $|\Sigma|$ outgoing neighbors — for
genomes $|\Sigma| = 4$

To navigate in the De Bruijn graph, we can simply
query all possible successors, and see which are
actually present.

Bloom Filters & De Bruijn Graphs

But, a Bloom filter still has false-positives, right?

May return some neighbors that are not actually present.

Pell et al., PNAS 2012, use a lossy Bloom filter directly

Chikhi & Rizk, WABI 2012, present a *lossless* data structure based on Bloom filters

Salikhov et al., WABI 2013 extend this work and introduce the concept of “cascading” Bloom filters

Pellow, Filippova & Kingsford, RECOMB 2016. Take advantage of “independence” of false positives to lower FP rate for Bloom Filter representations

First, some bounds

JOURNAL OF COMPUTATIONAL BIOLOGY
Volume 22, Number 5, 2015
© Mary Ann Liebert, Inc.
Pp. 336–352
DOI: 10.1089/cmb.2014.0160

Research Articles

On the Representation of De Bruijn Graphs

RAYAN CHIKHI,^{1,6} ANTOINE LIMASSET,³ SHAUN JACKMAN,⁴
JARED T. SIMPSON,⁵ and PAUL MEDVEDEV^{1,2,6}

We use the term membership data structure to refer to a way of representing a dBG and answering k -mer membership queries. We can view this as a pair of algorithms: (CONST, MEMB). The CONST algorithm takes a set of k -mers S (i.e., a dBG) and outputs a bit string. We call CONST a constructor, since it constructs a representation of a dBG. The MEMB algorithm takes as input a bit string and a k -mer x and outputs true or false. Intuitively, MEMB takes a representation of a dBG created by CONST and outputs whether a given k -mer is present. Formally, we require that for all $x \in \Sigma^k$, $\text{MEMB}(\text{CONST}(S), x)$ is true if and only if $x \in S$.

An NDS is a pair of algorithms, CONST and NBR. As before, CONST takes a set of k -mers and outputs a bit string. NBR takes a bit string and a k -mer and outputs a set of k -mers. The algorithms must satisfy that for every dBG S and a k -mer $x \in S$, $\text{NBR}(\text{CONST}(S), x) = \text{ext}(x) \cap S$. Note that if $x \notin S$, then the behavior of $\text{NBR}(\text{CONST}(S), x)$ is undefined. We observe that a membership data structure immediately implies an NDS because an NBR query can be reduced to eight MEMB queries.

In this section, we prove that a navigational data structure on de Bruijn graphs needs at least 3.24 bits per k -mer to represent the graph:

Theorem 1. *Consider an arbitrary NDS and let CONST be its constructor. For any $0 < \epsilon < 1$, there exists a k and $x \in \Sigma^k$ such that $|\text{CONST}(x)| \geq |x| \cdot (c - \epsilon)$, where $c = 8 - 3 \lg 3 \approx 3.25$.*

Representing the dBG exactly with an AMO

Chikhi and Rizk *Algorithms for Molecular Biology* 2013, **8**:22
<http://www.almob.org/content/8/1/22>



ALGORITHMS FOR
MOLECULAR BIOLOGY

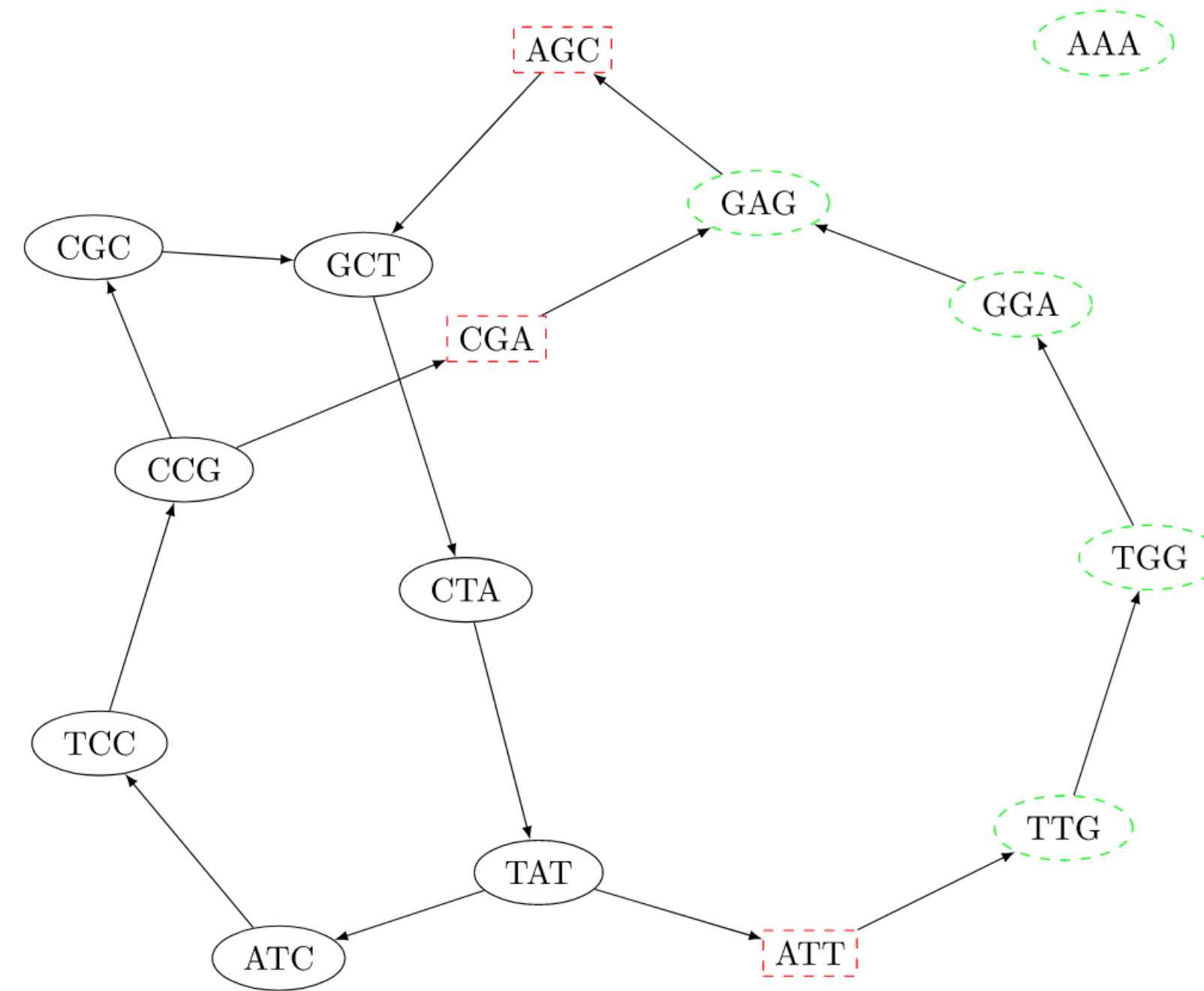
RESEARCH

Open Access

Space-efficient and exact de Bruijn graph representation based on a Bloom filter

Rayan Chikhi^{1*} and Guillaume Rizk²

Critical False Positives



(a)

**“toy” hash
func (not used in practice)**

$a_1 \dots a_k$	$\sum_{i=1}^k a_i^i \bmod 10$
ATC	0
CCG	0
TCC	5
CGC	6
...	...

(b)

Bloom filter

1
0
0
0
0
1
1
0
0
0

(c)

Nodes self-information:

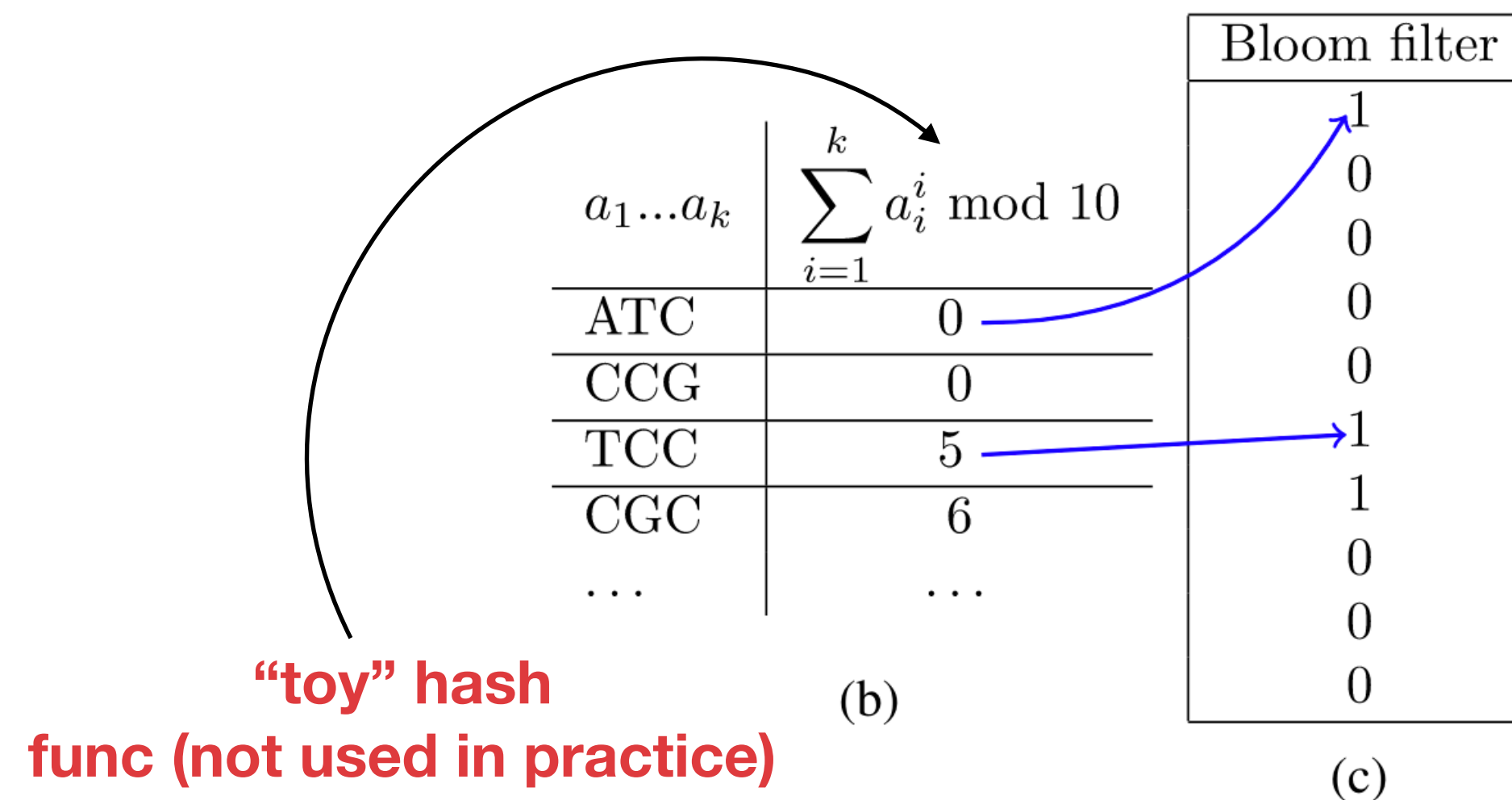
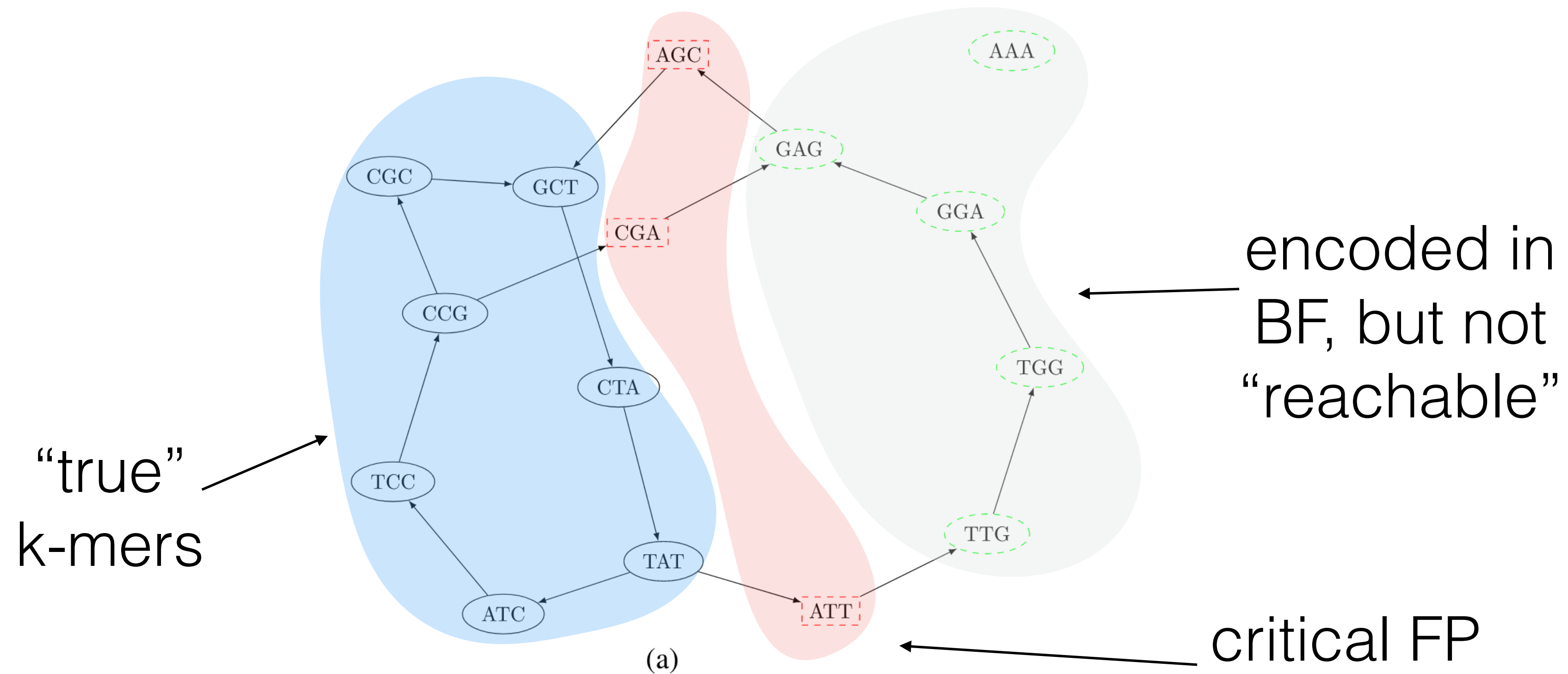
$$\lceil \log_2 \binom{4^3}{7} \rceil = 30 \text{ bits}$$

Structure size:

$$\underbrace{10}_{\text{Bloom}} + \underbrace{3 \cdot 6}_{\text{False positives}} = 28 \text{ bits}$$

(d)

Critical False Positives



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$$\lceil \log_2 \binom{4^3}{7} \rceil = 30 \text{ bits}$$

Structure size:

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(d)

Idea of Chkhi and Rizk

Assume we want to represent specific set T_0 of k -mers with a Bloom filter B_1

Key observation: in assembly, not all k -mers can be queried, only those having $k-1$ k -mers known to be in the graph.

The set T_1 of “critical false positives” (false neighbors of true k -mers) is *much* smaller than the set of all false positives and can be stored explicitly

Storing B_1 and T_1 is much more space efficient than other exact methods for storing T_0 . Membership of w in T_0 is tested by first querying B_1 , and if $w \in B_1$, check that it is *not* in T_1 .

Size of different components

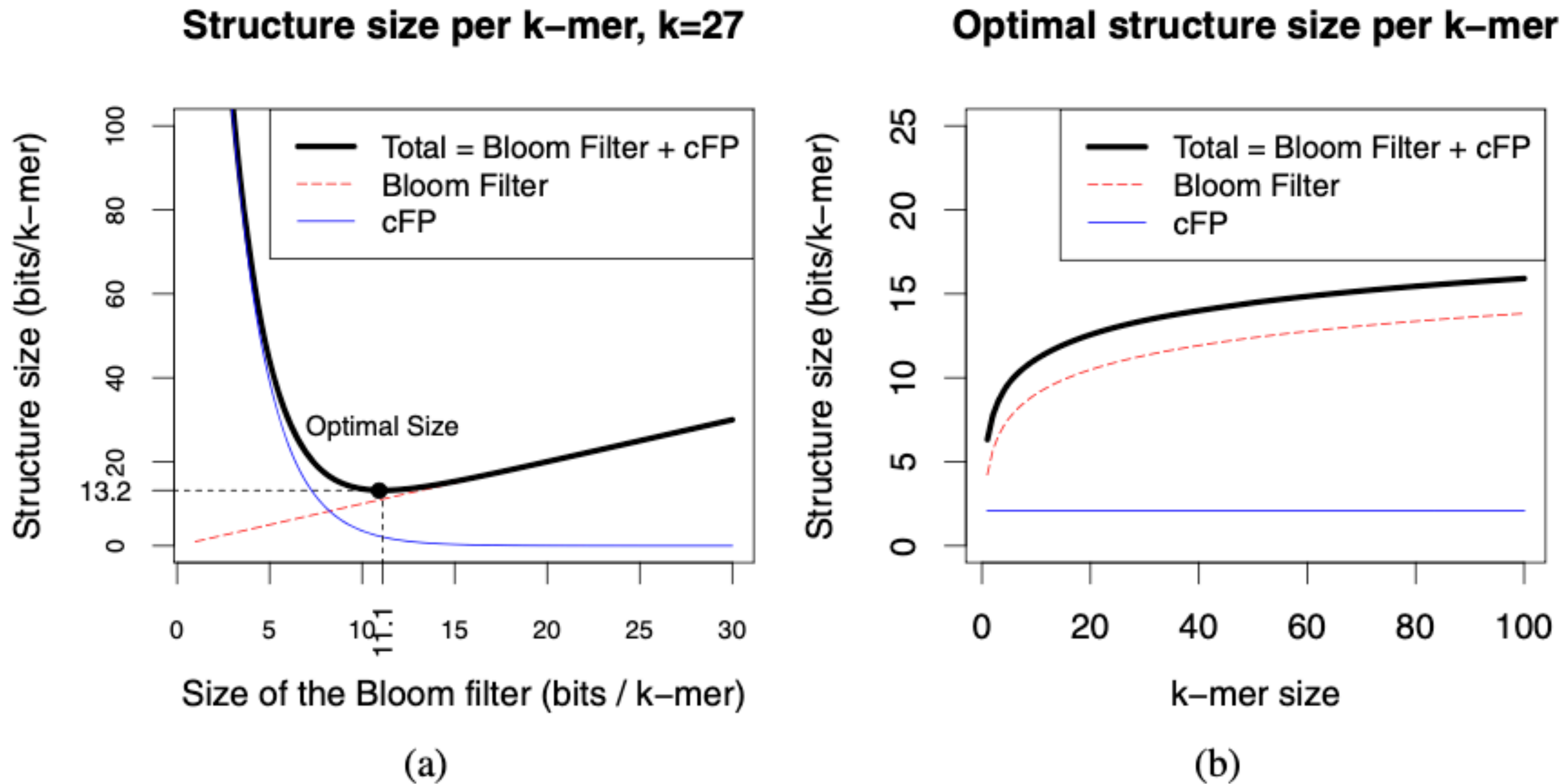


Figure 2 Optimal data structure size for the parameter r , then for the parameter k . (a) Structure size (Bloom filter, critical false positives) in function of the number of bits per k -mer allocated to the Bloom filter (also called ratio r) for $k = 27$. The trade-off that optimizes the total size is shown in dashed lines. (b) Optimal size of the structure for different values of k .

The cFP structure

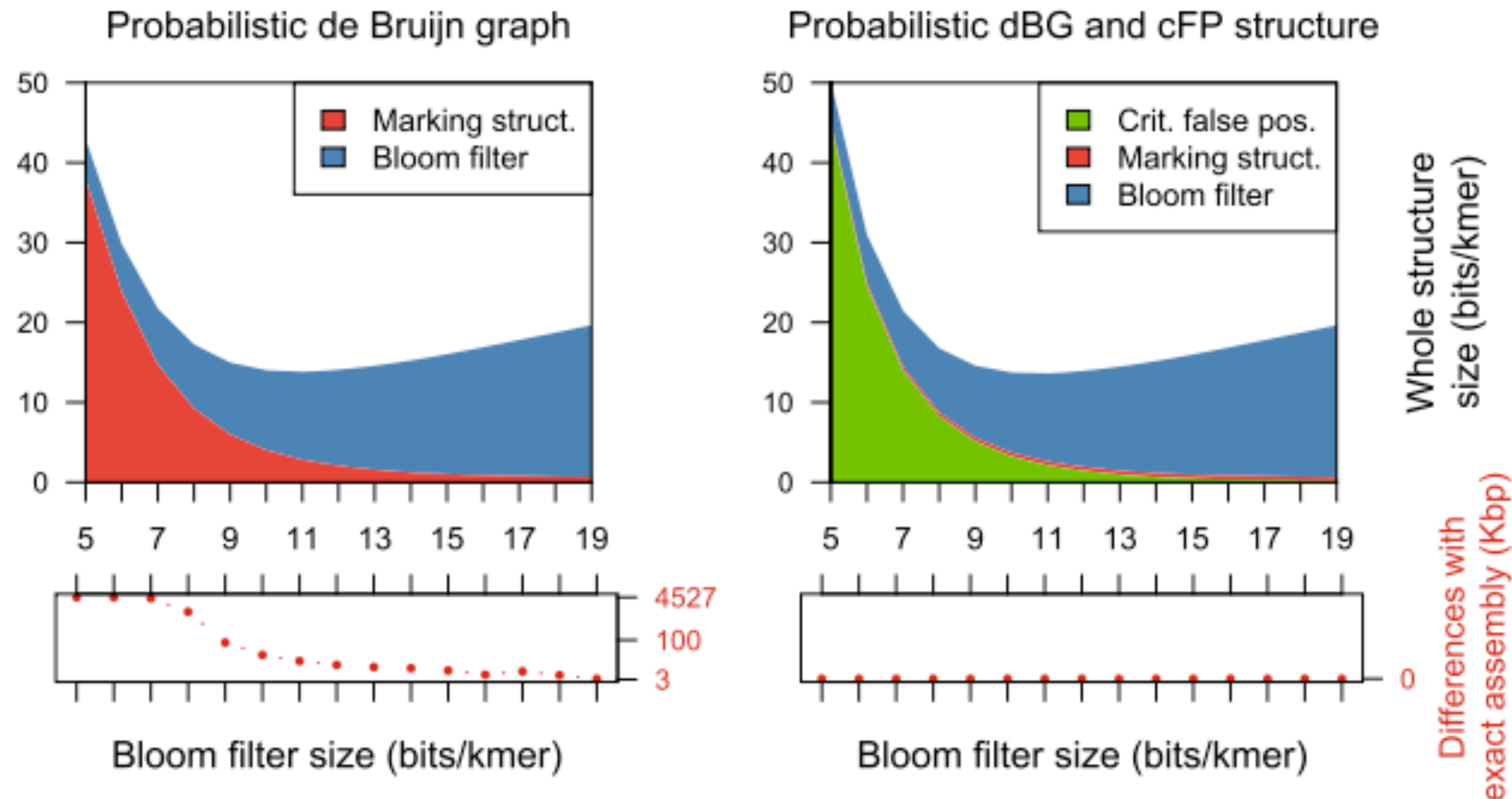


Figure 3 Data structure sizes for the probabilistic de Bruijn graph. Data structure sizes (Bloom filter, marking structure, and *cFP* if applicable) for the probabilistic de Bruijn graph with (top right) and without the *cFP* structure (top left), for an actual dataset (*E. coli*, $k = 23$). All plots are in function of the number of bits per k -mer allocated to the Bloom filter. Additionally, the difference is shown (bottom left and bottom right) between a reference assembly made using an exact de Bruijn graph, and an assembly made with each structure.

Human Genome Assembly on a Laptop!

Table 2 de novo human genome (NA18507) assemblies

Method	Minia	C. & B.	ABYSS	SOAPdenovo
Value of k chosen	27	27	27	25
Number of contigs (M)	3.49	7.69	4.35	-
Longest contig (kbp)	18.6	22.0	15.9	-
Contig N50 (bp)	1156	250	870	886
Sum (Gbp)	2.09	1.72	2.10	2.08
Nb of nodes/cores	1/1	1/8	21/168	1/16
Time (wall-clock, h)	23	50	15	33
Memory (sum of nodes, GB)	5.7	32	336	140

de novo human genome (NA18507) assemblies reported by our assembler (Minia), Conway and Bromage assembler [9], ABYSS [8], and SOAPdenovo [7]. Contigs shorter than 100 bp were discarded. Assemblies were made without any pairing information.

Efficiently enumerating cFP

Algorithm 1 Constant-memory enumeration of critical false positives

- 1: **Input:** The set $\mathcal{S}$ of all nodes in the graph, the Bloom filter constructed from $\mathcal{S}$, the maximum number M of elements in each partition (determines memory usage)
 - 2: **Output:** The set cFP
 - 3: Store on disk the set $\mathcal{P}$ of extensions of $\mathcal{S}$ for which the Bloom filter answers *yes*
 - 4: Free the Bloom filter from memory
 - 5: $D_0 \leftarrow \mathcal{P}$
 - 6: $i \leftarrow 0$
 - 7: **while** end of $\mathcal{S}$ is not reached **do**
 - 8: $P_i \leftarrow \emptyset$
 - 9: **while** $|P_i| < M$ **do**
 - 10: $P_i \leftarrow P_i \cup \{\text{next } k\text{-mer in } \mathcal{S}\}$
 - 11: **for each** k -mer m in D_i **do**
 - 12: **if** $m \notin P_i$ **then**
 - 13: $D_{i+1} \leftarrow D_{i+1} \cup \{m\}$
 - 14: Delete D_i, P_i
 - 15: $i \leftarrow i + 1$
 - 16: cFP $\leftarrow D_i$
-

Bloom filters & De Bruijn Graphs

So, we can make very small representation of the dBG.
But it's navigational! We can also make them:

Bioinformatics, 2018, 1–7
doi: 10.1093/bioinformatics/bty500
Advance Access Publication Date: 22 June 2018
Original Paper



Sequence analysis

Practical dynamic de Bruijn graphs

Victoria G. Crawford^{1,†}, Alan Kuhnle^{1,†}, Christina Boucher¹, Rayan Chikhi² and Travis Gagie^{3,*}

¹Department of Computer and Information Science and Engineering, University of Florida, Gainesville, FL 32306, USA, ²CNRS, CRISTAL, University of Lille, Lille, France and ³CeBiB and School of Computer Science and Engineering, Diego Portales University, Santiago, Chile

Dynamic &
membership

and even
weighted

Bioinformatics, 33, 2017, i133–i141
doi: 10.1093/bioinformatics/btx261
ISMB/ECCB 2017



deBGR: an efficient and near-exact representation of the weighted de Bruijn graph

Prashant Pandey¹, Michael A. Bender¹, Rob Johnson^{1,2} and Rob Patro^{1,*}

¹Department of Computer Science, Stony Brook University, Stony Brook, NY 11790, USA, ²VMWare, Inc., Palo Alto, CA 94304

*To whom correspondence should be addressed.

Succinct *k*-mer Sets Using Subset Rank Queries on the Spectral Burrows-Wheeler Transform *

Jarno N. Alanko, Simon J. Puglisi, Jaakko Vuohtoniemi

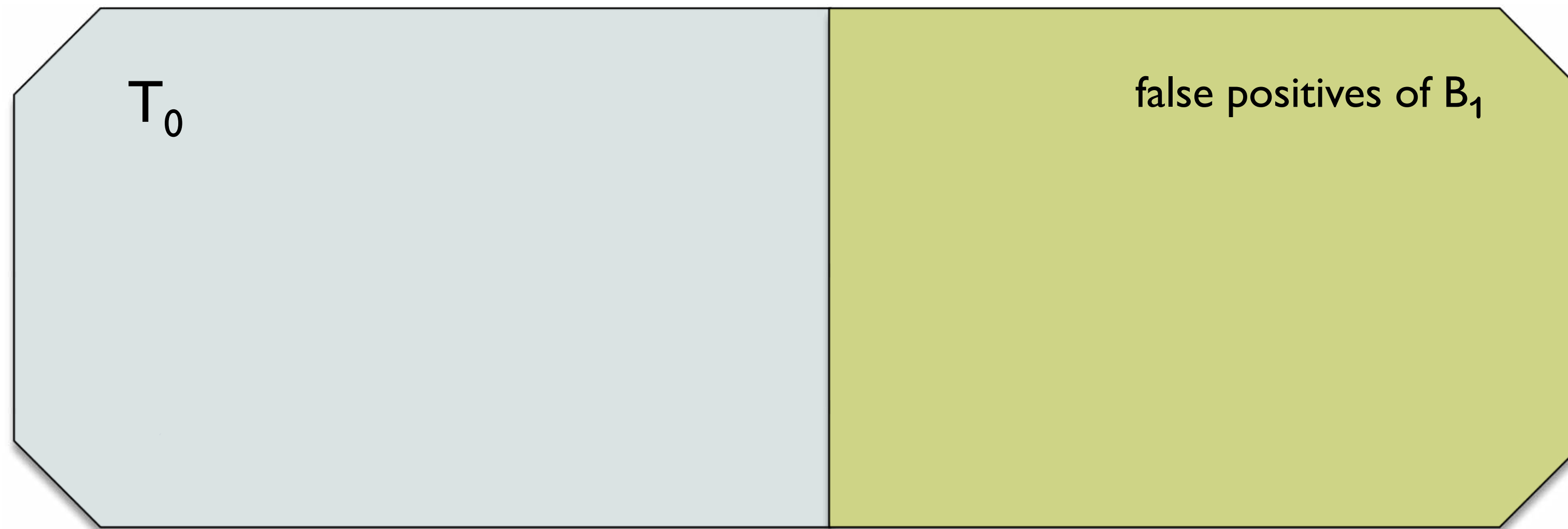
doi: <https://doi.org/10.1101/2022.05.19.492613>

This article is a preprint and has not been certified by peer review [what does this mean?].



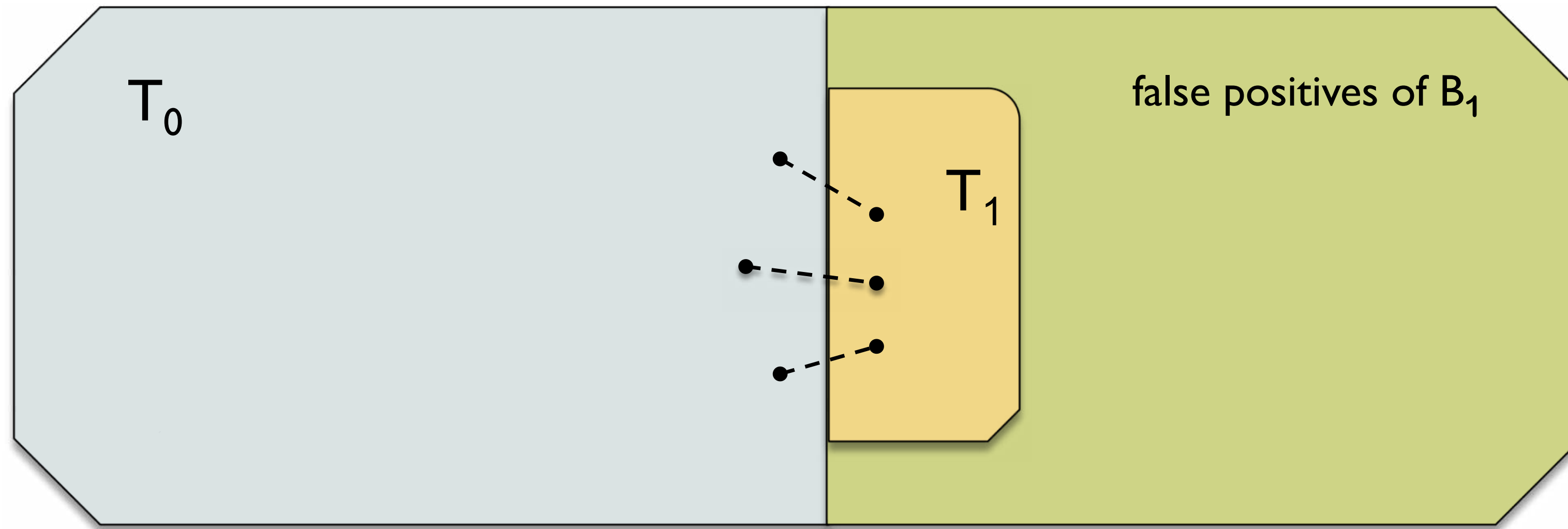
Current static membership
state-of-the-art for (non-compacted)
de Bruijn Graphs

Bonus Slides



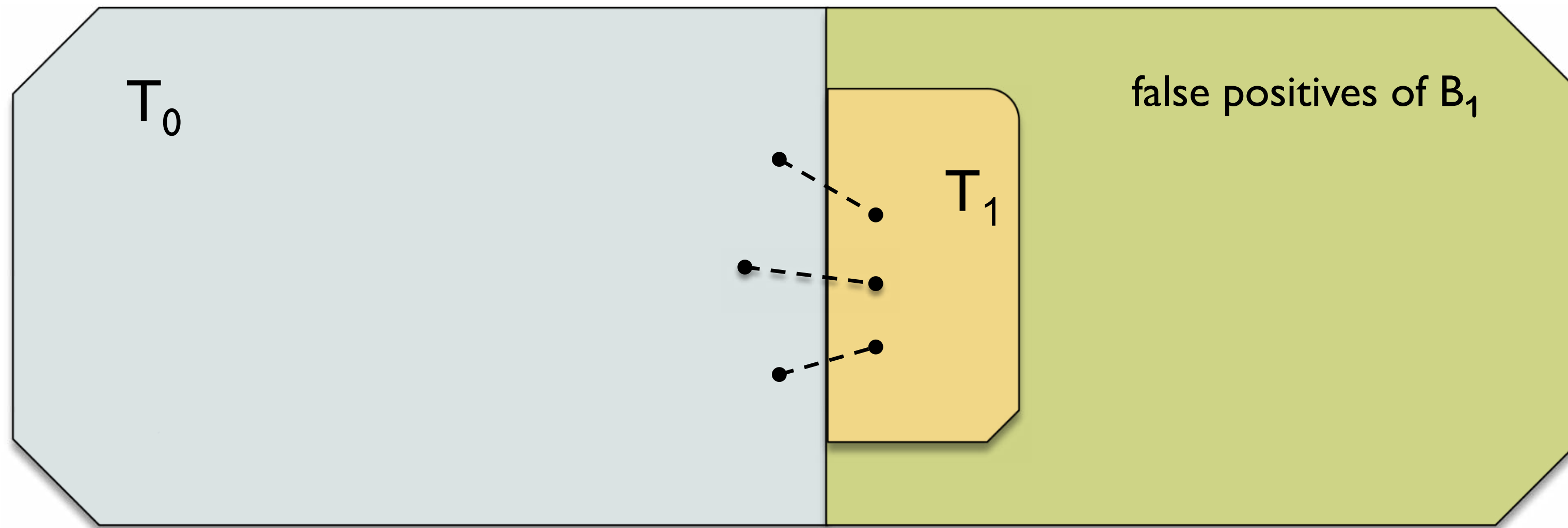
- ▶ Represent T_0 by Bloom filter B_1





- ▶ Represent T_0 by Bloom filter B_1
- ▶ Compute T_1 ('critical false positives') and represent it e.g. by a hash table





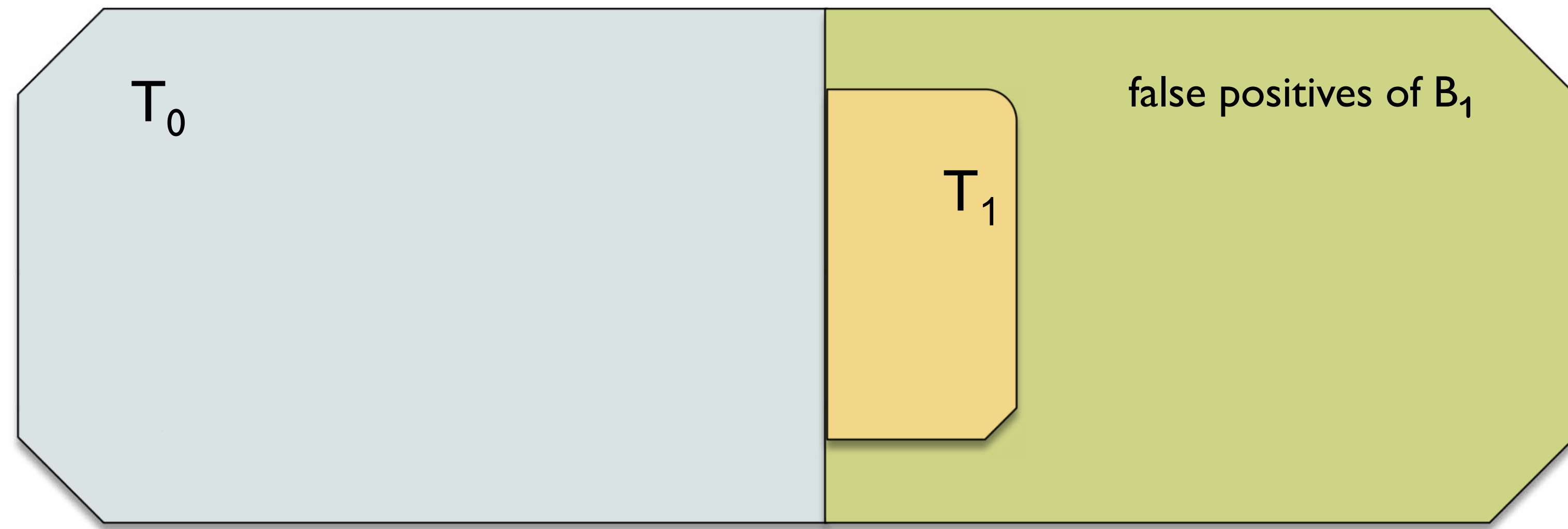
- ▶ Represent T_0 by Bloom filter B_1
- ▶ Compute T_1 ('critical false positives') and represent it e.g. by a hash table
- ▶ Result (example): 13.2 bits/node for $k=27$ (of which 11.1 bits for B_1 and 2.1 bits for T_1)

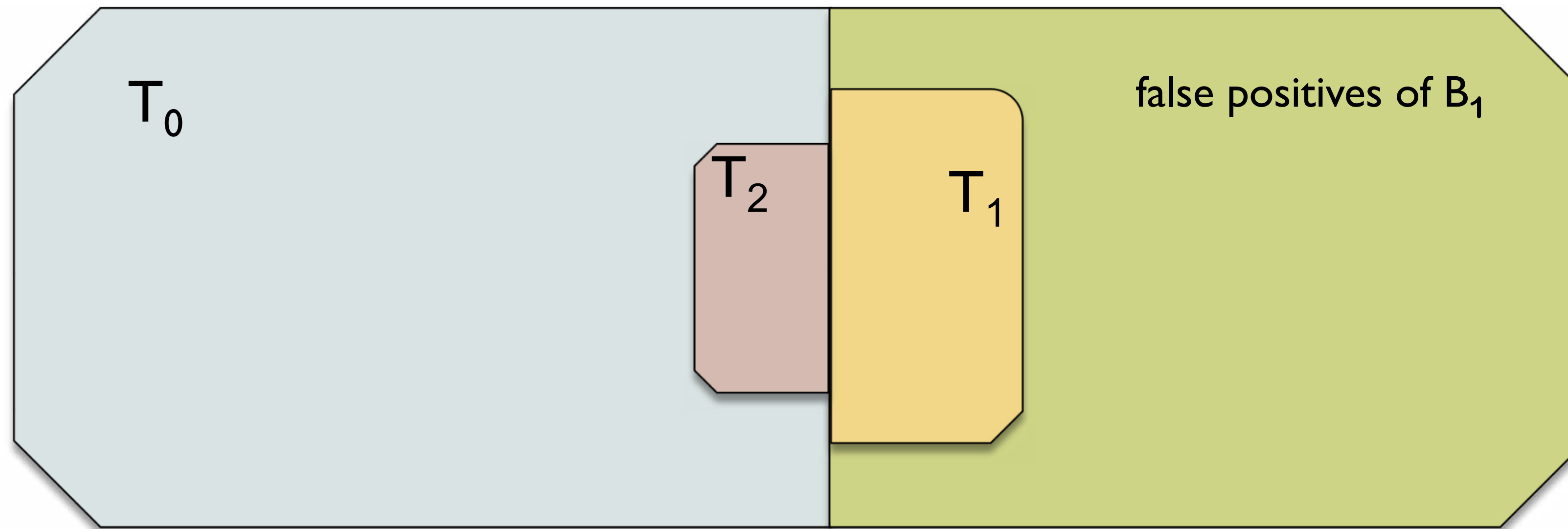


Improving on Chikhi and Rizk's method

- ▶ *Main idea*: iteratively apply the same construction to T_1 i.e. encode T_1 by a Bloom filter B_2 and set of 'false-false positives' T_2 , then apply this to T_2 etc.
- ▶  *cascading Bloom filters*

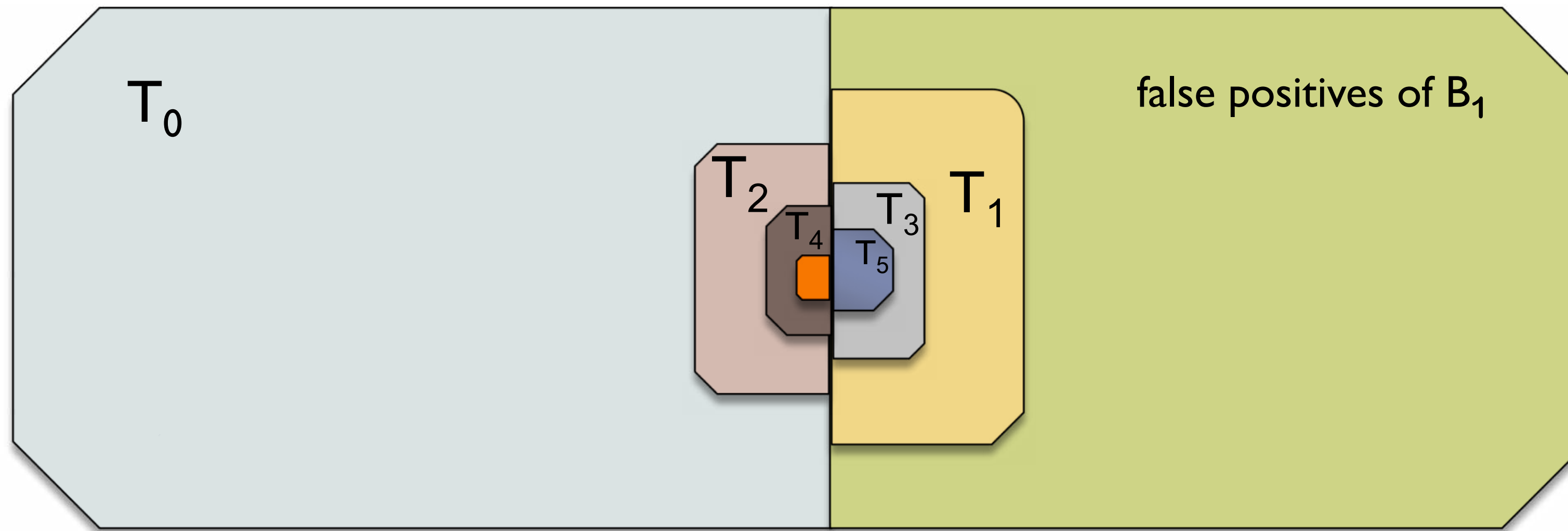






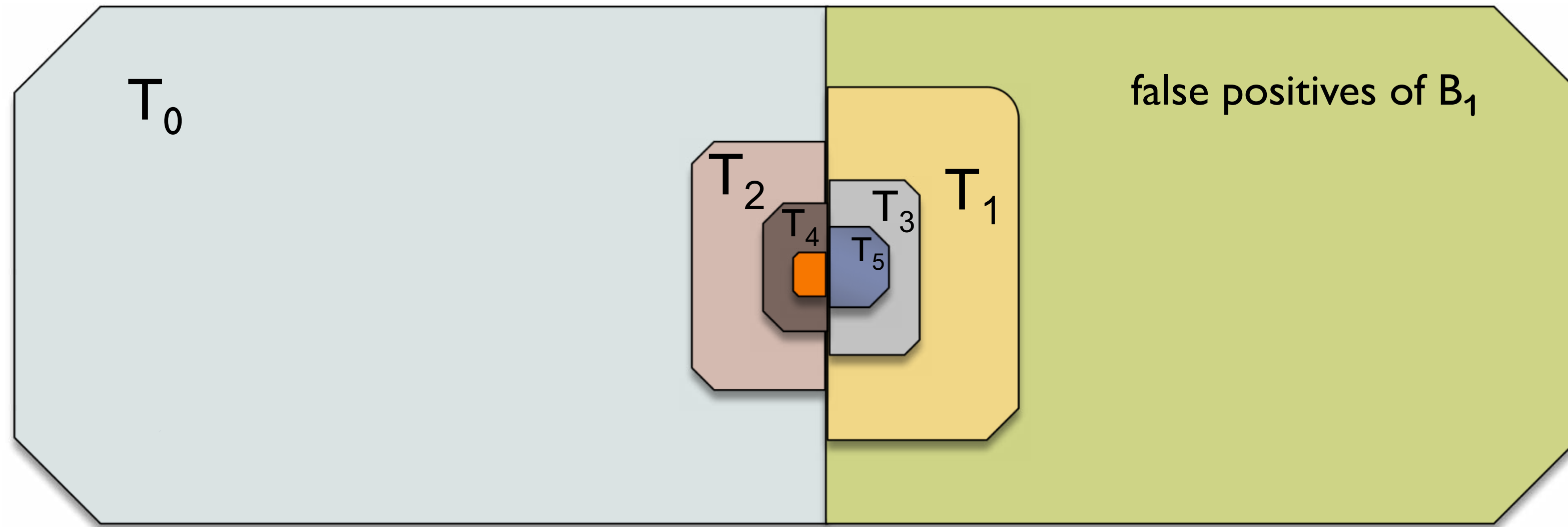
- ▶ further encode T_1 via a Bloom filter B_2 and set T_2 , where $T_2 \subseteq T_0$ is the set of k -mers stored in B_2 by mistake ('false² positives')





- ▶ further encode T_1 via a Bloom filter B_2 and set T_2 , where $T_2 \subseteq T_0$ is the set of k -mers stored in B_2 by mistake ('false² positives')
- ▶ iterate the construction on T_2
- ▶ we obtain a sequence of sets $T_0, T_1, T_2, T_3, \dots$ encode by Bloom filters $B_1, B_2, B_3, B_4, \dots$ respectively
- ▶ $T_0 \supseteq T_2 \supseteq T_4 \supseteq \dots, T_1 \supseteq T_3 \supseteq T_5 \supseteq$





Lemma [correctness]: For a k -mer w , consider the smallest i such that $w \notin B_{i+1}$. Then $w \in T_0$ if i is odd and $w \notin T_0$ if i is even.

- ▶ if $w \notin B_1$ then $w \notin T_0$
- ▶ if $w \in B_1$, but $w \notin B_2$ then $w \in T_0$
- ▶ if $w \in B_1$, $w \in B_2$, but $w \notin B_3$ then $w \notin T_0$
- ▶ etc.



Assuming infinite number of filters

Let $N=|T_0|$ and $r=m_i/n_i$ is the same for every B_i . Then the total size is

$$\underbrace{rN}_{|B_1|} + \underbrace{6rNc^r}_{|B_2|} + \underbrace{rNc^r}_{|B_3|} + \underbrace{6rNc^{2r}}_{|B_4|} + \underbrace{rNc^{2r}}_{|B_5|} + \dots = N(1+6c^r) \frac{r}{1-c^r}$$

The minimum is achieved for $r=5.464$, which yields the memory consumption of **8.45** bits/node



Infinity difficult to deal with ;)

- In practice we will store only a small finite number of filters $B_1, B_2, \dots, B_t$ together with the set T_t stored explicitly
- $t=1 \Rightarrow$ Chkhi&Rizk's method
- The estimation should be adjusted, optimal value of r has to be updated, example for $t=4$

k	optimal r	bits per k -mer
16	5.776737	8.555654
32	6.048557	8.664086
64	6.398529	8.824496
128	6.819496	9.045435

Table: Estimations for $t=4$. Optimal r and corresponding memory consumption



Compared to Chikhi&Rizk's method

k	“Optimal” (infinite) Cascading Bloom Filter	Cascading Bloom Filter with $t = 4$	Data structure of Chikhi & Rizk
16	8.45	8.555654	12.0785
32	8.45	8.664086	13.5185
64	8.45	8.824496	14.9585
128	8.45	9.045435	16.3985

Table: Space (bits/node) compared to Chikhi&Rizk
for $t=4$ and different values of k .



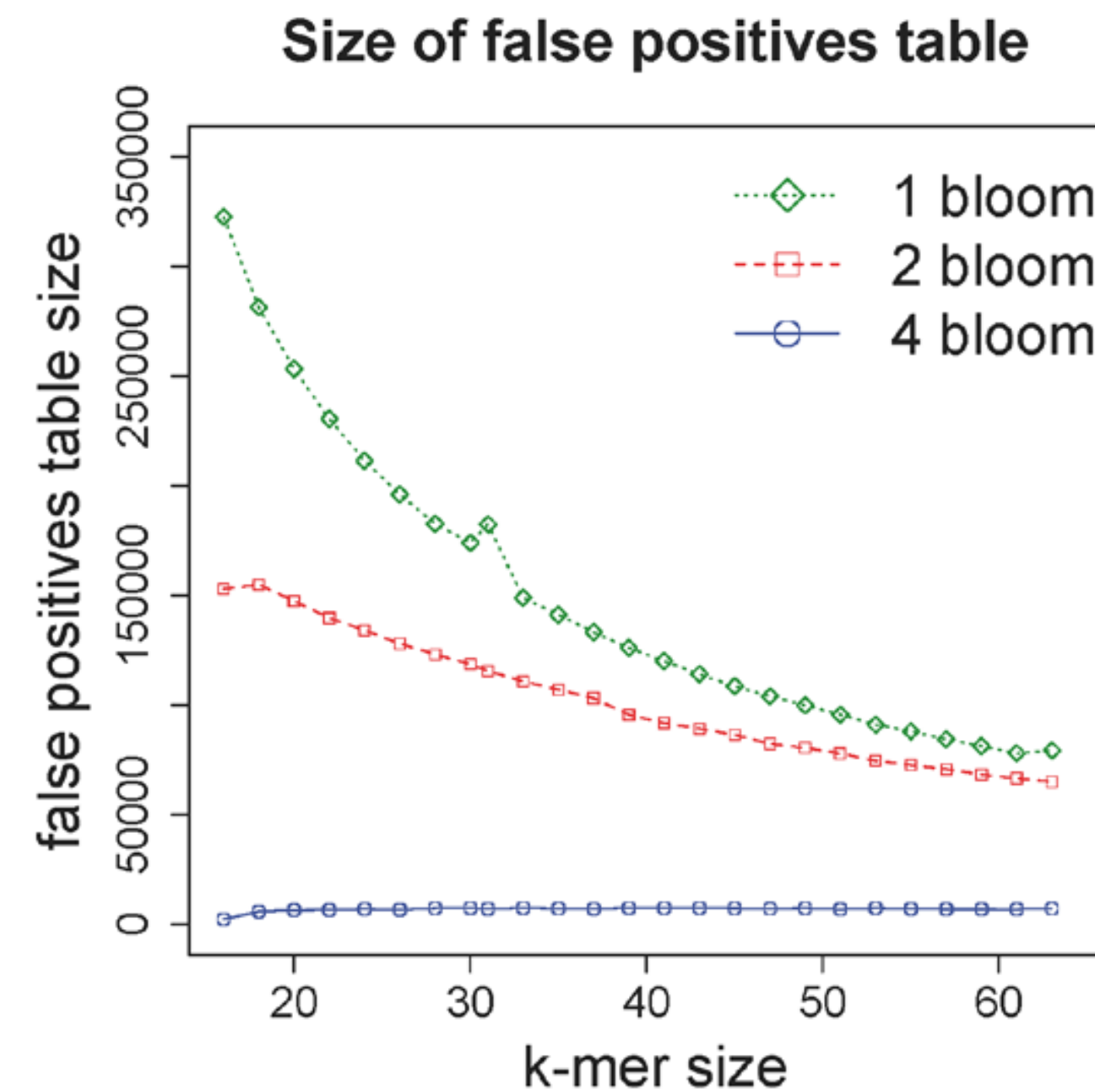
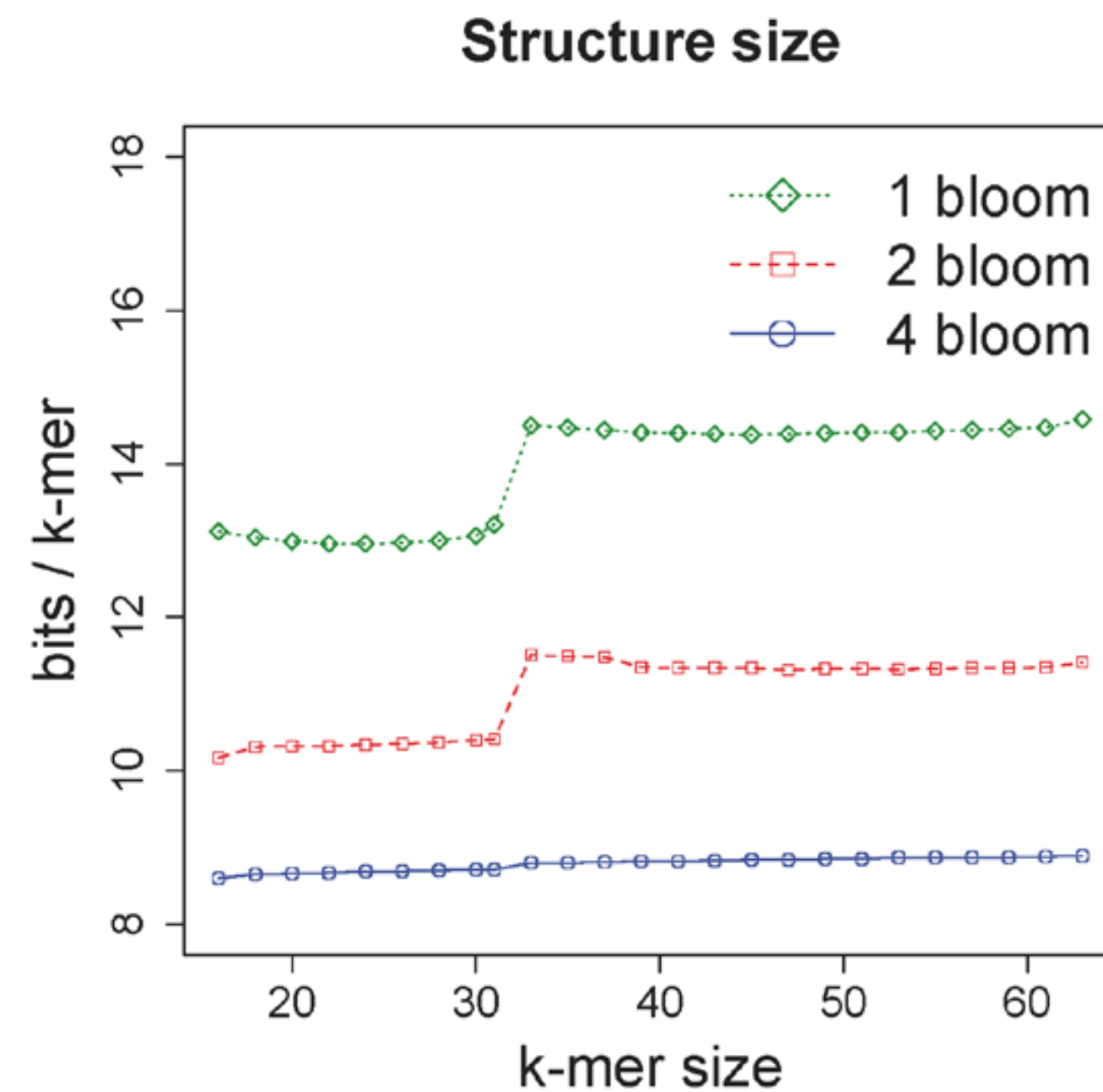
We can cut down a bit more ...

- Rather than using the same r for all filters $B_1, B_2, \dots$, we can use different properly chosen coefficients $r_1, r_2, \dots$
- This allows saving another 0.2 – 0.4 bits/k-mer



Experiments I: E.Coli, varying k

- 10M E.Coli reads of 100bp
- 3 versions compared: 1 Bloom (=Chikhi&Rizk), 2 Bloom ($t=2$) and 4 Bloom ($t=4$)



Experiments II: Human dataset

- 564M Human reads of 100bp (~17X coverage)

Method	1 Bloom	2 Bloom	4 Bloom
Construction time (s)	40160.7	43362.8	44300.7
Traversal time (s)	46596.5	35909.3	34177.2
r (bits)	11.10	8.10	6.56
Bloom filters size (MB)	$B_1 = 3250.95$	$B_1 = 2372.51$ $B_2 = 292.65$	$B_1 = 1921.20$ $B_2 = 496.92$ $B_3 = 83.39$ $B_4 = 21.57$
False positive table size (MB)	$T_1 = 545.94$	$T_2 = 370.96$	$T_4 = 24.07$
Total size (MB)	3796.89	2524.12	2547.15
Size (bits/k-mer)	12.96	10.37	8.70

Experiments I (cont)

